Tahseel AI

Project Documentation

DEPI Graduation Project: an agentic credit-control copilot for Egyptian B2B distributors

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Repository: https://github.com/alialsharqawi/Tahseel-Copilot

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Project Planning and Management

This part answers the LMS requirement to clarify the idea, goals, and scope; set a timeline; distribute tasks and responsibilities; identify risks and how to handle them; and define success metrics. It is assembled from the governed sources: the scope (`docs/foundation/Tahseel_AI_Final_Concept.md`), the product specification (`docs/technical/Tahseel_AI_PRD.md`), the business analysis (`docs/business/Tahseel_AI_Business_Strength.md`, specifically its validation plan and risk sections), the backlog (`docs/team/backlog.csv`, 50 tasks), and the team decisions (`docs/governance/Tahseel_AI_Decision_Log.md`, in particular D29, D34, D37, and D38). When this part summarizes them, the source file is the owner of the detail (D35).

An honest note on the plan itself. The milestones below are our proposal, built from the backlog's dependency waves. The binding calendar is the one the instructor sets on the LMS; the first submission (the three dynamic links) went live on 4 November 2026 as confirmed in `PROJECT_STATUS.md`. Team availability (real hours per week per member) is collected in the skills matrix at onboarding, and the plan is revised when those numbers exist.

1. The idea and its goals

One paragraph. Tahseel AI is a credit-control and receivables assistant for mid-sized Egyptian B2B distributors. It reads the company's own receivables from Excel/ERP exports, tells the finance team **who to call today, why, and what to say** based on payment history and the company's own credit policy, drafts the message, and **records the outcome** so the next day's advice gets better. Every collection action stays under human approval; the system advises at the desk and never contacts a customer by itself.

Goals.

1. **Product goal:** move the owner or finance manager from "we chase everyone and forget half" to a daily, explained, policy-grounded action list. Success on the core demo path: aging, explanation, drafting, approval gate, and outcome recording all work end to end.
2. **Learning goal:** demonstrate a complete AI-assisted product workflow — deterministic business logic, retrieval with citations (D9/D21), an agent with tool calls (D12), memory, and a human approval gate (D36) — on realistic synthetic data (D20), with honest measurement.
3. **Business goal (evaluated, not blocking):** the thesis "late follow-up, not customer hostility, is a meaningful share of Egyptian B2B lateness" (hypothesis A8), and the capstone retirement plan for a real-world value discussion.

2. Scope: what is in, what can be cut

The scope is structured evidence, not a wish list (from D29 and the Decision Log, section E):

Tier	Meaning	Examples
Core (41 tasks)	Never cut	Receivables engine with tests, risk explanations, policy RAG with citations, the approval gate, the audit log, memory (notes, promises, outcomes), tenant isolation with its test, the basic evaluation with the ablation
Should (7 tasks)	Built after the core slice works, in the cut order	Forecast, Health Report, draft check, reminder cap, local-competitor scan

Tier	Meaning	Examples
Delivery (2 tasks)	Scheduled after the core slice works	Deployment, backups-style Operational tasks

If time runs short we cut in the fixed order of Decision Log section E (production polish first, Arabic polish beyond the three core screens, guardrail extras, forecast sophistication, older-note embeddings, Health Report extras) — never the Core list. This is our main schedule-protection mechanism and it is decided in advance, not during a panic.

Out of scope for the capstone (from the Final Concept and PRD): automatic contacting of customers by the agent, automatic dispute status changes, late fees or interest computation, replacing the ERP or accounting system, multi-tenant SaaS onboarding.

3. Team, roles, and responsibilities

Five members. Names and agreed responsibilities are being finalized at the onboarding kickoff (they will be recorded here and in `TEAM.md`; anything the team agrees on is committed, not only discussed).

Area of ownership	Approx. share of backlog	Skills code	Notes
Team Leader — Ali Al-Sharqawi	LMS deliverables, governance, review, integration	GIT, DOC	Submits on the LMS (confirmed); holds the review/approval gate (D37)
Databases and data (ERD, migrations, synthetic data)	~9 tasks	DB, DATA	—
Backend and AI (FastAPI, engine, retrieval, agent)	~18 tasks	API, AI	strongest-skill pairing for the core engine
Frontend (Next.js bilingual dashboards)	~12 tasks	UI, FRONT	Arabic-first RTL (D6)
Docs, QA, and demo materials	~11 tasks	DOC, QA	evidence labels, evaluation runs, slides

The distribution rule (D37): tasks are assigned by the skills matrix (`docs/team/skills.csv`) using `tools/team_tools.py suggest`, then a person takes the decision, then any override is recorded. One task = one issue = one branch = one pull request; every task has an assignee and a **different** reviewer; a person holds at most two active tasks. Sensitive tasks (security, tenancy, authentication, the approval gate, financial formulas) get a reviewer who can do that skill alone. Learning tasks are allowed with a stronger reviewer. Branch names follow `type/T-id-short-description` (D37, amended), and every member commits under their own account — the collaboration record is part of what is graded (D38).

AI use policy (D38): AI is a tutor, not a substitute — each member must be able to explain what they submit; AI-use disclosures appear in pull requests.

4. Work plan and timeline

The backlog's dependency waves give the real shape of the plan (sizes: S ≈ half a day, M ≈ 1–2 days, L ≈ 3–5 days; totals from `python tools/team_tools.py backlog`):

Phase	Approx. calendar (proposal)	Waves	Content	Load
P0 — Kickoff and setup	5–8 Nov 2026	Wave 1	Board, labels, protection, skills matrix, first PR per member (T-001, T-022 partially)	~9 person-days
P1 — Plan and documents	9–15 Nov 2026	Wave 2 (this part, T-090; issue assembly)	Project plan, assignments, issues in all waves	~2 person-days
P2 — Sprint 1: data and groundwork	16 Nov → 4 Dec 2026	Waves 3–4	Docker (T-003), CI (T-004), model/migrations (T-010), synthetic data (T-020), auth/roles (T-011), engine (T-013)	~16 person-days
P3 — Sprint 2: the AI slice	7–25 Dec 2026	Waves 5–6	Retrieval with citations, agent tools and memory, the approval gate, dashboards for aging and the daily queue	~56 person-days
P4 — Sprint 3: UI, guardrails, polish	4–22 Jan 2027	Waves 7–8	Bilingual UI, guards, forecast, Health Report, production basics	~29 person-days
P5 — Evaluation and delivery	Jan 2027 (instructor calendar)	Waves 9–10	Evaluation + ablation, slides, demo video, and the final PDF via `tools/build_docs_pdf.py` (T-091)	~3 person-days

Milestone gates (checked at the end of each phase, evidence in the repo):

1. **Gate 1 (P1):** all 50 issues exist on the board; every member has ≥ 1 merged PR; skills matrix filled.
2. **Gate 2 (P2):** the engine runs against the synthetic dataset; the 19-table schema is migrated; CI green on every PR; the aging tests pass.
3. **Gate 3 (P3):** the 12-step demo path works end to end on the core: aging → explanation with citations → draft → approval → recorded outcome.
4. **Gate 4 (P4):** the cut decision exercised (if needed) with the Core intact; the basic evaluation executed.
5. **Gate 5 (P5):** all LMS deliverables replaced in place (the links stay live); final presentation by the confirmed Team Leader.

The cut order (Decision Log section E) applies at any gate where the remaining work exceeds the remaining time.

Why dates are honest here: the wave totals sum to ~115 person-days of estimated work for a five-person team working part-time in parallel; if the real availability from the skills matrix is materially lower, phases shrink by cutting Should/Delivery per the agreed order, **not** by compressing the Core.

5. Risks and how we handle them

Risk	Impact	Likelihood	Mitigation	Evidence label
Part-time availability collapses around exams	Schedule slips	High	Cut order in advance (D29/section E); at most 2 active tasks per person; weekly 15-min sync; the 48-hour rule (a member silent 2 days gets asked, not blamed)	Plan of record

Risk	Impact	Likelihood	Mitigation	Evidence label
No real distributor data	The demo rests on synthetic data	Certain (constraint)	Synthetic generator with realistic patterns (D20), stated as synthetic everywhere (D27); the validation plan probes the real world after the capstone	Sourced (constraint)
LLM/model choice underperforms on Arabic	Explanations degrade	Medium	Model choice deferred until a tool-calling + Arabic test (D12); fallbacks defined in the Decision Log	To verify
Embedding quality for mixed Arabic/English notes	Retrieval misses	Medium	Embedding model chosen after the Arabic retrieval test (D16); hybrid retrieval (BM25 + vectors) so keyword hits survive embedding weakness	To verify
RAG retrieval error could mislead a rule	Wrong advice → trust damage	Medium	Two-layer design: thresholds evaluated by the deterministic engine, RAG only explains and cites (D21); approval gate keeps every action human (D36)	Sourced (design)
Global tools "do it too" / competitor pressure	Positioning weakens	Low to medium	Positioning is Arabic + Excel-first + policy citations + human approval — verified gap (assumption A9, scan task T-094 rechecks it)	To verify
A member stalls or drops a task	Dependency blocks	Medium	One task = one branch; two-task cap; the leader reassigns, recorded on the issue; nothing invisible (the audit log, D37)	Plan of record
Grade risk: claims outrun evidence	Rubric penalties	Medium (mitigated)	Evidence labels on every claim (D19); banned-wording rule D27; the pitch is a demo-voice script only within D39's rails; numbers come from code	Sourced (process)
Key concentration: the Team Leader holds governance	Slow if he is unavailable	Medium	Each member merges small PRs after review; the Decision Log records overrides; the weekend demo script can be run by any pair with the recorded scenario	To verify

6. Success metrics

For the capstone (D34-scoring alignment with the rubric): the graded dimensions are architecture, UI, RAG and memory, code quality, presentation, real-world value, and collaboration (10%). The concrete, checkable targets:

1. **Demo integrity:** the 12-step demo path succeeds end to end, and the numbers shown on screen are the engine's numbers (aging, exposure, risk level, priority ranking are exactly what the deterministic engine computes — verified by unit tests).
2. **Explanation quality:** every policy-based recommendation shows its rule and its source citation; every agent recommendation shows tool calls and costs (D8/D36 vocabulary).
3. **Evaluation (basic, never cut):** ~15 agent cases, ~10 RAG queries (including no-policy and conflicting-policy cases), ~5 safety tests, plus the ablation. We report actual results **including weaknesses** (D29) — the results section is written after the runs, never before.
4. **Collaboration:** every commit under its own human account (D38); review trail visible; each member's first PR merged in week 1 (see §4, Gate 1).

5. **Documentation integrity:** the LMS link suite stays live; the PDF regenerates from source (T-091); the hierarchy rule (D35) holds — nothing new above is contradicted by anything below.

For the product after the capstone (from the Business Strength validation plan, we report them as plans, not results): 10 distributor conversations test A8 and the daily-list value; 2–3 free receivables health pilots measure the candidate outcome: days beyond terms moved and collection effort saved; then a pricing test. Kill criteria are written in `Tahseel_AI_Business_Strength.md` §8.

Literature Review

This part surveys the solutions that already exist for collecting overdue invoices, says what each does well and where it falls short, compares them with Tahseel AI, and records what we learned from the research behind our design.

1. Method and limits

We read vendor pages, official documentation, press releases, and software marketplace listings, and a few research papers. All web sources were accessed in early November 2026 (the sources of the first competitor pass, 31 October 2026, are listed in the business file).

Limits, stated plainly - We did not install or test any product. Everything about a product comes from what its maker or a listing says. - Vendor pages are marketing. Figures such as "reduces past due by 20%" are **vendor claims**, and we do not repeat them as facts. - Marketplace listings (Capterra, GetApp, Xero) are secondary sources. - "Not seen" means we did not find it on the pages we read. It is **not** proof that a feature does not exist. - The scan is not exhaustive, and the local (Egyptian and regional) scan is the weakest part. The business file keeps a proper local scan as an open task (assumption A9).

Evidence labels (Glossary): **Sourced** (the source is cited and its caveats stated), **Indicative** (a secondary or marketing source), **To verify** (not checked).

2. The solution landscape

2.1 Spreadsheets and manual follow-up

The real competitor. It is free and familiar. It gives no ranking and no explanation, and the history of promises and calls lives in people's heads. This is the situation Tahseel is built to replace.

2.2 Reminders inside accounting and ERP software

Odoo follow-ups. Odoo's documentation describes follow-up levels that trigger actions according to the number of days an invoice is overdue. The actions include an email, an SMS, a letter, and a manual action that creates a task. A follow-up report lists the overdue invoices, a responsible person can be assigned per customer, and reminders can be automatic or manual. The documentation also has a section on a debtor's trust level, which we did not read in detail. An Odoo partner page states that an Egyptian localization package with accounts related to the Egyptian Tax Authority exists. *Sourced (official documentation); the Egypt detail is Indicative.*

Daftra. A cloud accounting and ERP product used in Egypt and the region. Its pages describe e-invoicing for the Egyptian Tax Authority, tracking of payment status, automatic reminders by email or SMS, and installment management, in Arabic and English. We did not see risk scoring, prioritization, or an AI assistant. *Indicative (vendor pages).*

Strength: the data is already there, and local compliance and Arabic are handled. **Weakness:** these are rule-based reminders by days overdue. They send notices; they do not rank customers, explain why, or ground a recommendation in the company's own policy (not seen).

2.3 Collections automation for small and medium businesses

ezyCollect (now part of Sidetrade) is an add-on for accounting systems such as MYOB, Xero, NetSuite, and Dynamics 365. It offers automated reminders, payment options, and credit risk monitoring, and its maker announced an autonomous AI collections agent for small businesses. *Sourced (vendor pages and a trade report).*

CreditorWatch Collect (formerly Debtor Daddy) works with Xero and other accounting tools. Its help page describes using credit risk scores to prioritize high-risk customers and tracking call outcomes. *Sourced (vendor help page).*

Chaser connects to Xero, QuickBooks, NetSuite, and others. Listings describe reminders by email, SMS, and automated call, AI insights on which debtors to prioritize, and cash-flow forecasting. The listing we saw shows English as the supported language. *Indicative (marketplace listings).*

Upflow is a B2B receivables platform with reminders, workflows, and analytics. Listings show English and French. *Indicative.*

Strength: mature workflows, integrations with popular accounting software, and in several cases risk-based prioritization and forecasting. **Weakness for our setting:** they are built around Western accounting ecosystems and, as far as we saw, English-language communication. They assume the data lives in one of those systems.

2.4 Enterprise order-to-cash suites and agentic AI

HighRadius sells a collections module to large companies. Its pages describe an AI-prioritized worklist built from more than 20 variables and 12 months of payment history, a suggested action for each account, assignment to collectors by their capacity, a prediction of delinquency 30 days ahead, automated dunning, an in-app dialer, and more than 50 ERP integrations. It claims large gains (for example 20% lower past due and 30% higher collector productivity); these are **vendor claims**. It also says that implementation and the return on investment arrive within three to six months (also a vendor claim). *Sourced (vendor pages).*

Sidetrade offers an order-to-cash platform with an AI agent called Aimie. A press release describes an agent that places first-contact and follow-up calls and leaves voicemails on its own. A press report says Sidetrade deployed Aimie for Accor's finance teams across the Middle East, Africa, and Asia-Pacific. *Sourced (press release); the regional deployment is Indicative (one report).*

Gaviti targets medium and large enterprises with AI-driven prioritization, dispute tracking, cash application, credit management, and ERP integrations. *Sourced (vendor page) for prioritization; the rest Indicative.*

Strength: depth, data, and scale. **Weakness for our setting:** cost and integration effort suit large companies; the direction of travel is toward **autonomous** collection, where the AI contacts the customer without a person approving each message.

3. Comparison

"Not seen" means not found on the pages we read.

Solution	Segment	Ranks accounts	Forecast	Contacts customers	Arabic or Egypt	Policy-grounded recommendation
Excel and manual	Everyone	No	No	A person	n/a	No
Odoo follow-ups	ERP users	By days overdue (levels)	Not seen	Email, SMS, letter (rule-driven)	Egyptian localization exists	Not seen
Daftra	Egypt and region	Not seen	Not seen	Email, SMS reminders	Arabic and English; e-invoicing	Not seen
ezyCollect	Small and medium	Not seen	Not seen	Automated, plus an AI agent	Not seen	Not seen
CreditorWatch Collect	Small and medium	Yes (risk scores)	Not seen	Automated	Not seen	Not seen
Chaser	Small and medium	Yes (AI insights)	Yes	Email, SMS, call	English listed	Not seen
Upflow	B2B teams	Not seen	Not seen	Automated	English, French listed	Not seen
Gaviti	Enterprise	Yes	Not seen	Not checked	Not checked	Not seen
HighRadius	Enterprise	Yes (with a suggested action)	Predicts delinquency	Automated dunning, dialer	Not checked	Not seen
Sidetrade (Aimie)	Enterprise	Yes	Not checked	Autonomous calls and emails	Regional deployment reported	Not seen
Tahseel AI	**Egyptian distributors**	**Yes, with every factor shown**	**Yes (Should tier), with a range**	**Never; a person sends**	**Arabic and English, Arabic drafts**	**Yes: rules with sources, plus cited passages**

4. What we learned

1. **A ranked worklist with a suggested action is standard.** Several tools already do it. Our queue is therefore not our edge; it is the minimum. What we add is the reasoning behind it: the factors, the policy rule, and the memory used.
2. **The industry is moving toward autonomy.** Sidetrade and others now sell agents that call customers themselves. Tahseel makes the opposite choice on purpose: the AI recommends and drafts, a person approves, and nothing is sent by the system. We do not claim this is better. It trades speed for control and trust, and whether customers prefer it is a hypothesis we test (Business_Strength, section 6).
3. **Local reminders exist but are rule-based.** Odoo and Daftra send reminders by days overdue and handle Arabic and local compliance. We did not see prioritization or policy-grounded advice there. That gap is a hypothesis (A9), not a finding.
4. **Enterprise tools need integration time.** One vendor gives three to six months for implementation and return on investment (a vendor claim). Starting from an Excel file, with no integration, is our wedge, and it is also a hypothesis.

5. **Vendor outcome numbers are marketing.** We make no outcome claim for Tahseel until we have real data.
6. **Explainability and policy citations were not visible** on the vendor pages we read. Whether the tools offer them is to verify.

5. Research foundations

Source	What it says (paraphrased)	How it shapes Tahseel
Lewis et al., *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*, NeurIPS 2020	Combining a language model with a retrievable external memory helps with two weaknesses of models that store knowledge only in their parameters: giving provenance for answers and updating knowledge	Policy RAG with citations; policies live in documents that the company can update (D9)
Ji et al., *Survey of Hallucination in Natural Language Generation*, arXiv 2022	Generated text can be fluent and still unfaithful to its source; the survey reviews how to measure and reduce it	Numbers come from tools, not from the model; we measure numeric errors against an LLM-only baseline (D5, D11)
Rudin, *Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead*, Nature Machine Intelligence, 2019	For high-stakes decisions, models that are interpretable from the start are preferable to explanations added to a black box	A transparent weighted-rule risk score with every factor listed, instead of an opaque model (D8)

We do not claim that these papers validate our product. They support the design choices; our own evaluation has to show that the choices work.

6. Where Tahseel stands

Where we are alike	Where we choose differently	What we have not shown
A ranked list with a suggested action; automated calculation; drafts of messages	Human approval by default; Arabic first; a start from an Excel file; recommendations grounded in the company's own policy with citations; factor-level explanations	That anyone prefers these choices; that our recommendations are good; that the product shortens lateness

The assumptions behind the differences are in the business file (A8 and A9), with the tests that could weaken them.

7. Open points

- Verify whether the tools above offer policy citations or factor-level explanations.
- Run a proper scan of Egyptian and regional tools (Google, LinkedIn, ERP partners, app directories).
- Search the academic literature on payment-delay prediction. We did not, and our risk score is deliberately rule-based, not predictive.

8. Sources

Accessed in early November 2026 unless noted.

Enterprise suites - HighRadius, collections management: https://www.highradius.com/products/collections-management-software/ - HighRadius, worklist variables: https://www.highradius.com/?p=195740 - HighRadius, AI-based worklist prioritization: https://www.highradius.com/software/autonomous-receivables/collections-management/ai-based-worklist-prioritization/ - Sidetrade press release (AI cash collection agent), via StreetInsider: https://www.streetinsider.com/Globe+Newswire/Sidetrade+introduces+first-ever+AI+Cash+Collection+Agent%2C+autonomous+and+interactive/24833370.html - Sidetrade at Accor, MarketScreener: https://www.marketscreener.com/news/sidetrade-deploys-autonomous-ai-agent-for-accor-s-finance-function-ce7e51dcdf8ef724 - ezyCollect launches Aimie, ChannelLife: https://channellife.com.au/story/ezycollect-launches-aimie-to-tackle-late-sme-payments - Gaviti: https://gaviti.com/?p=6362

Small and medium business tools (first pass, 31 October 2026) - ezyCollect: https://ezycollect.io/?p=21250445 ; CreditorWatch Collect: https://help.creditorwatch.com.au/en/articles/11650655-what-is-creditorwatch-collect ; Chaser (listing): https://www.capterra.ae/reviews/157101/chaser ; Upflow (listing): https://www.capterra.com.au/software/193097/upflow

Accounting and ERP - Odoo follow-up documentation (documentation repository): https://github.com/odoo/documentation/pull/4497/files - Odoo Egyptian localization (partner page): https://techscopeco.odoo.com/en/accounting - Daftra, sales and invoicing: https://daftra.com/en/sales ; e-invoicing in Egypt: https://daftra.com/en/egy-electronic-invoice

Research - Lewis et al., arXiv:2005.11401: https://arxiv.org/abs/2005.11401 - Ji et al., arXiv:2202.03629: https://arxiv.org/abs/2202.03629 - Rudin, Nature Machine Intelligence 1(5), 206-215: https://doi.org/10.1038/s42256-019-0048-x

Market evidence (first pass, 31 October 2026): see `Business_Strength`, section 11 (Allianz Trade and the FCIB survey).

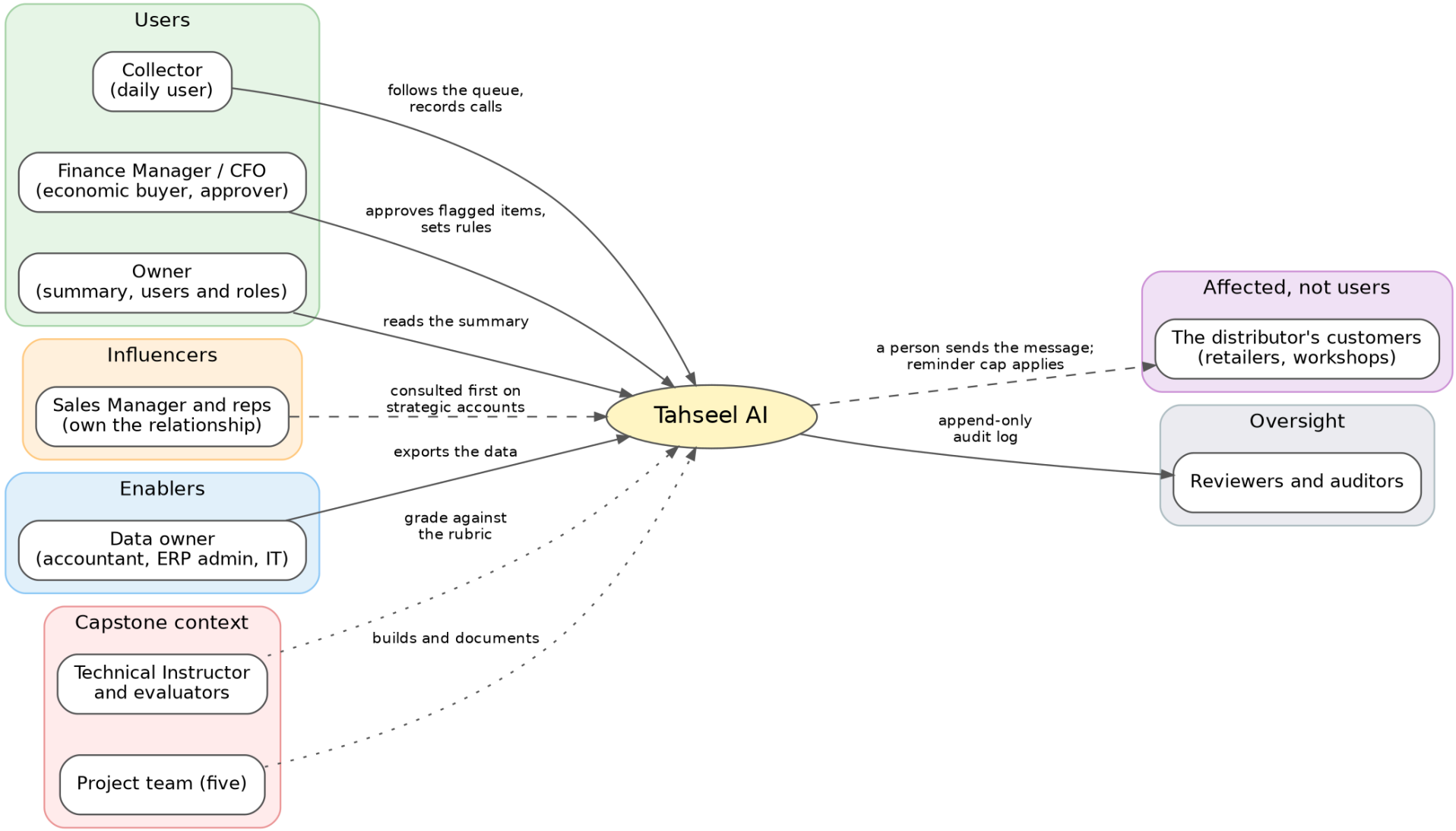
Requirements Gathering and Stakeholder Analysis

This part identifies who is affected by Tahseel AI and what they need, then states what the system must do (functional requirements) and how well it must do it (quality requirements). It is built from the product specification (`docs/technical/Tahseel_AI_PRD.md`), the scope (`docs/foundation/Tahseel_AI_Final_Concept.md`), and the business analysis (`docs/business/Tahseel_AI_Business_Strength.md`).

An honest note on the evidence. We have not interviewed any distributor. Every stakeholder need below is a **hypothesis** from desk research, to be tested in the validation plan. The requirements are therefore what we decided to build, not findings from users.

1. Stakeholder analysis

1.1 Stakeholder map


Users
Collector (daily user)
Finance Manager / CFO (economic buyer, approver)
Owner (summary, users and roles)
Influencers
Sales Manager and reps (own the relationship)
Enablers
Data owner (accountant, ERP admin, IT)
Capstone context
Technical Instructor and evaluators
Project team (five)
follows the queue, records calls
approves flagged items, sets rules
reads the summary
consulted first on strategic accounts
exports the data
grade against the rubric
builds and documents
Tahseel AI
a person sends the message; reminder cap applies
append-only audit log
Affected, not users
The distributor's customers (retailers, workshops)
Oversight
Reviewers and auditors

1.2 Stakeholders, needs, and concerns

Influence and interest are our estimates (Hypothesis).

Stakeholder	Relation to the product	Needs	Concerns	Influence	Interest
Finance Manager / CFO	Economic buyer and champion; the Finance Manager role	Lower days beyond terms; a lower overdue share; cash visibility for 7 and 30 days; control over the collection team	Cannot verify what the AI says; losing control; wrong numbers	High	High
Owner	Beneficiary (also the buyer in smaller firms); the Owner role	An understandable summary; to know which accounts threaten cash	Reports that arrive late or lack explanation	High	Medium
Collector (collection officer)	Daily user; the Collector role	To know who to contact first; history and the relevant policy in one place; professional messages quickly	Scattered information; more work instead of less	Medium	High
Sales Manager and reps	Influencers, not product users in the MVP	Protection of relationships with key customers	A collection call that damages a key account	Medium to high (can block)	Medium
Data owner (accountant, ERP admin, IT)	Enabler	A simple export of data without disrupting the ERP	Messy exports; risk to the live system	Medium	Low to medium
The distributor's customers (retailers, workshops)	Affected, not users	Good supply and fair terms	Pressure at a bad moment; too many reminders	Low (indirect)	Low
Reviewers and auditors	Read the audit log	Evidence of who approved what	Untraceable decisions	Low	Medium
Technical Instructor and evaluators (capstone)	Judge the project against the rubric	Architecture, UI, RAG and memory, code quality, presentation, real-world value, collaboration	Unsupported claims; a demo that does not work	High (for the grade)	High
Project team (five)	Builds and documents	A clear scope and ownership	Scope growth; unclear roles	High	High

1.3 From needs to requirements

Stakeholder need	Requirements that answer it
Know who to chase first, and why	FR-06, FR-07, FR-08, FR-31
Trust the numbers	FR-04, FR-05, FR-19, FR-21, NFR-04, NFR-06
Trust the policy statements	FR-10, FR-11, FR-12
Stay in control of what is sent	FR-23, FR-25, FR-26, FR-27, NFR-05
Do not repeat work or forget promises	FR-14, FR-15, FR-16, FR-17
Protect key relationships	FR-09, FR-28, FR-29

Stakeholder need	Requirements that answer it
See cash and risk at a glance	FR-03, FR-05, FR-30, FR-31
Start from the files they already have	FR-01, FR-02
Keep data safe and separate	FR-34, FR-35, NFR-01, NFR-02, NFR-03
Use it in Arabic	FR-23, FR-32, NFR-10
Show evidence of who approved what	FR-27, NFR-05

2. User stories

From the specification, section 9, with the additions made by later decisions.

Finance Manager: see total outstanding and overdue receivables; see high-risk customers; see expected cash; understand why a customer is high risk; approve flagged recommendations; confirm policy rules and set tenant settings.

Collector: see a prioritized Collection Queue; see customer history before calling; see the recommended next action; generate a message draft; record the result of a call; record a promise to pay; mark an invoice as disputed.

Owner: know whether collection is improving; know which accounts threaten cash flow; get an understandable summary; manage users and roles.

3. Functional requirements

Tiers (Decision Log, D29): **Core** = Must (the demo cannot run without it; never cut), **Should** = Should (built after the core slice, in the cut order), **Delivery** = scheduled after the core slice works. **Capability** refers to `Final_Concept`, section 3. **Test** refers to the acceptance criteria in the PRD, section 30.

Import and receivables

ID	Requirement	Tier	Capability	Test
FR-01	The user can upload a CSV or Excel file. The system proposes a column mapping from the headers and masked sample rows, the user confirms it, the system validates the records, and a preview is shown before anything is imported	Core	C1	AC-001
FR-02	The import creates or updates customers and invoices and records payments, including partial payments. Duplicate invoice numbers and missing due dates are reported with fix-or-skip choices	Core	C1	AC-001
FR-03	After the first upload the system shows a Receivables Health Report: aging, risk distribution, the top 20 accounts to chase, and expected cash	Should	S3	Demo checklist
FR-04	The system computes outstanding amounts, days overdue, and aging per invoice and per customer. Overdue is computed from the due date and is never stored. Payments are the source of truth for what has been paid	Core	C2	AC-002
FR-05	The system computes portfolio KPIs: totals, the overdue ratio, average days beyond terms, and DSO	Core	C2	AC-002

Risk, priority, and the queue

ID	Requirement	Tier	Capability	Test
FR-06	Every customer has a risk score from 0 to 100, built from payment behavior with default weights (days overdue 30, historical delay 25, broken promises 25, recent behavior 20). The score lists each factor and its points, and the points add up to the score	Core	C3	AC-003, AC-010
FR-07	Every customer has a priority score = risk x overdue exposure x urgency, scaled so the top account is 100. The three factors are shown	Core	C4	AC-003
FR-08	The Collection Queue ranks customers with overdue, non-disputed exposure. Each row shows a suggested action computed by rules, without an LLM	Core	C4	Demo checklist
FR-09	For a strategic customer, the suggested action and the recommendation are to talk to the sales rep first	Should	S2	Demo checklist

Policy

ID	Requirement	Tier	Capability	Test
FR-10	Users can upload policy documents. Retrieval returns passages with the document and section as a citation, and the tenant filter is applied inside the query	Core	C5	AC-004, AC-007
FR-11	When no relevant policy is found the system says so, and when policies conflict it flags the case for human review. It never invents policy	Core	C5	AC-004
FR-12	Structured policy rules (thresholds, grace periods, escalation triggers) are evaluated by code and shown with their source document and section. A Finance Manager confirms them. They are entered manually or loaded by a seed script	Should	S5	AC-011
FR-13	Weights, caps, grace periods, escalation triggers, the reminder cap, and tone defaults are tenant settings, not code. Changes are audited	Should	S6	AC-010, AC-011

Memory

ID	Requirement	Tier	Capability	Test
FR-14	A Collector can record a note, a call result, and a promise to pay. Each entry carries author, timestamp, and source	Core	C7	AC-009
FR-15	The status of a promise (pending, kept, broken) is derived by rule from payments. It feeds the risk score and the urgency	Core	C6	AC-009
FR-16	The agent reads customer memory and cites it. The interface shows which memories were used	Core	C6	AC-009
FR-17	The agent keeps the conversation context so follow-up questions work. Conversation memory is never a source of facts, and nothing moves into customer memory unless a person saves it	Core	C6	AC-009
FR-18	A Finance Manager or Owner can edit or delete a memory entry. A deletion is a soft delete and is written to the audit log	Core	C6	AC-006

Agent

ID	Requirement	Tier	Capability	Test
FR-19	The agent answers questions about a customer by calling tools. Every number in an answer comes from a tool result, and a post-check confirms it	Core	C8	Agent eval
FR-20	The agent recommends a next action with reasons and citations. A new recommendation starts as pending	Core	C8	AC-004, AC-005
FR-21	Every agent run records the model, the tools used, token counts, cost, and latency	Core	C8	Agent eval
FR-22	Retrieved text, notes, and uploaded documents are treated as data. No tool runs because a document says so	Core	C8	Safety tests

Drafts, approval, and audit

ID	Requirement	Tier	Capability	Test
FR-23	The system generates message drafts in Arabic with tone options: formal, friendly, and firm. The default tone follows the customer tier	Core	C9	Demo checklist
FR-24	A rule-based check flags drafts that contain legal threats or forbidden phrases	Should	S7	Safety tests
FR-25	A user can approve, edit, or reject a recommendation. The original and edited draft text are stored with the reason. The API rejects any execution of a recommendation that is not approved	Core	C10	AC-005
FR-26	A recommendation flagged by a policy rule needs a Finance Manager or Owner. A Collector can send it for manager review	Core	C10, C13	AC-014
FR-27	A task is created only from an approved recommendation. Every recommendation (including ones that are never approved), approval, edit, rejection, and tool call is written to an append-only audit log	Core	C11	AC-006

Guardrails and forecast

ID	Requirement	Tier	Capability	Test
FR-28	A person can set an invoice to disputed. It leaves the ranked queue, appears in a Disputed list, follow-up pauses, and the manager is alerted. Only a manager role can clear the status	Should	S7	AC-012
FR-29	A configurable cap limits reminders per customer per period. Beyond the cap, a draft needs a manager override	Should	S7	AC-013
FR-30	The system shows expected cash for 7 and 30 days with a low, expected, and high range and a confidence label. It never presents the figure as guaranteed	Should	S1	AC-008

Interface, access, and delivery

ID	Requirement	Tier	Capability	Test
FR-31	The interface has the Executive dashboard, the Collection Queue, and the Customer Profile with the AI panel, plus the import screen	Core	C12	Demo checklist
FR-32	The interface works fully in Arabic (right to left) and in English, with a language toggle. All text comes from translation dictionaries	Should	S4	Demo checklist

ID	Requirement	Tier	Capability	Test
FR-33	The AI panel shows designed states for "no relevant policy", "policies conflict", "invoice disputed", and "reminder cap reached"	Should	S5, S7	AC-004, AC-012, AC-013
FR-34	Every record belongs to a tenant. A tenant cannot read another tenant's data, including through vector search	Core	C13	AC-007
FR-35	There are three roles (Owner, Finance Manager, Collector), enforced on the server. A Collector sees only assigned customers. Only the Owner manages users and roles	Core	C13	AC-014
FR-36	A basic evaluation (about 15 agent cases, about 10 retrieval queries, about 5 safety tests) and the comparison of an LLM given the raw file against the tool-grounded system are run, and the real results are reported	Core	C14	Evaluation report
FR-37	The system runs with Docker Compose, environment-based configuration, migrations, a seed script, a health-check endpoint, and CI that runs lint and tests. A public HTTPS deployment follows once the core slice works	Delivery	P1	CI

4. Quality requirements

The notice asks for quality requirements such as performance, security, and ease of use. Items marked **target** are goals to measure; they are not results.

ID	Category	Requirement	How we verify it	Tier
NFR-01	Security	Tenant isolation and server-side role checks. A cross-tenant test returns nothing; a Collector cannot reach unassigned customers	Automated tests (AC-007, AC-014)	Core
NFR-02	Security	TLS in transit; encryption at rest as a deployment setting; hashed passwords; secrets only in environment variables; a least-privilege database user	Deployment checklist; secret scan in CI	Core / Delivery
NFR-03	Privacy	Minimal personal data in prompts. Only synthetic data in the demo. Before real data is used, providers are restricted and their retention settings reviewed	Prompt review; documented procedure	Core
NFR-04	Accuracy	Every number in an agent answer matches a tool result. Numeric errors are measured against an LLM-only baseline and reported	Agent eval and the ablation	Core
NFR-05	Auditability	The audit log is append-only and the application role cannot update or delete it	Database grants; AC-006	Core
NFR-06	Explainability	Every score lists factors and points; every policy statement has a source or says none was found	AC-003, AC-004, AC-010	Core
NFR-07	Performance (target)	The queue and the dashboard make no LLM call. On the demo dataset the queue loads in about 2 seconds, and an agent answer arrives in about 30 seconds while showing which tool it is calling	Timed runs on the demo dataset	Should
NFR-08	Reliability	LLM calls have timeouts, retries, and a fallback model. A failure shows a plain message and leaves nothing half-saved	Failure-injection test	Should
NFR-09	Ease of use (target)	From the queue, a Collector reaches a draft in at most 3 clicks, and team members complete the main flow without instructions	Walkthrough with team members	Should

ID	Category	Requirement	How we verify it	Tier
NFR-10	Localization and accessibility	Arabic (right to left) and English; Western digits; `Intl` formats; text contrast at WCAG AA; color never the only signal; every action reachable by keyboard	Review against the UI checklist	Should
NFR-11	Maintainability and code quality	Typed schemas, modules by responsibility, unit tests for the calculations, CI, a README with setup, and no secrets in the repository	CI results; code review	Core
NFR-12	Configurability	Weights, thresholds, and rules are data, not code	AC-010, AC-011	Should
NFR-13	Portability	The whole system starts with Docker Compose on a clean machine	A clean-machine run	Delivery
NFR-14	Cost control	No LLM call on page loads; per-run token and cost logging; a spending limit at the gateway	Run logs	Core
NFR-15	Honesty	The demo is labeled as synthetic data; no claim of real customers, pilots, or measured accuracy that we do not have	Wording rules (Decision Log, D27)	Core

5. Constraints and assumptions

Constraints - A team of five, working to the capstone schedule, with an agreed cut order (Decision Log, section E). - No access to real distributors. All data is synthetic. - LLM access goes through a third-party gateway (OpenRouter). - Documentation is in English; the product interface is in Arabic and English. - The stack is fixed by the decisions: Next.js, FastAPI, PostgreSQL with pgvector, Docker Compose.

Assumptions (to validate; see the validation plan) - Distributors can export their receivables to Excel (A1). - Many companies have a documented credit policy, or can write one (A3). - A meaningful share of lateness comes from late or inconsistent follow-up (A8). - Collectors will act on suggestions that come with reasons (A5).

6. Out of scope for the capstone

Payments and wallets; automatic credit blocking; automatic WhatsApp or email sending; legal advice; ERP and CRM integrations (only the CSV and Excel adapter is built); a trained ML risk model; credit-limit recommendations. Later ideas are in the roadmap in `Final_Concept`, section 12.

7. Prioritization

Tier	Meaning	Functional requirements
Core (Must)	The demo cannot run without it. Never cut	26
Should	Built after the core slice, cut in the order in the Decision Log, section E	10
Delivery (Could)	Scheduled after the core slice works; a recorded demo is the fallback	1

8. Validation of these requirements

The requirements reflect decisions, not user research. The plan to test the underlying assumptions is in `Business_Strength`, section 8: ten interviews with finance and collection people, Health Check pilots on anonymized data, and a pricing test. Each assumption has a test and a kill criterion. Requirements will be

revised when real users contradict a hypothesis.

System Architecture

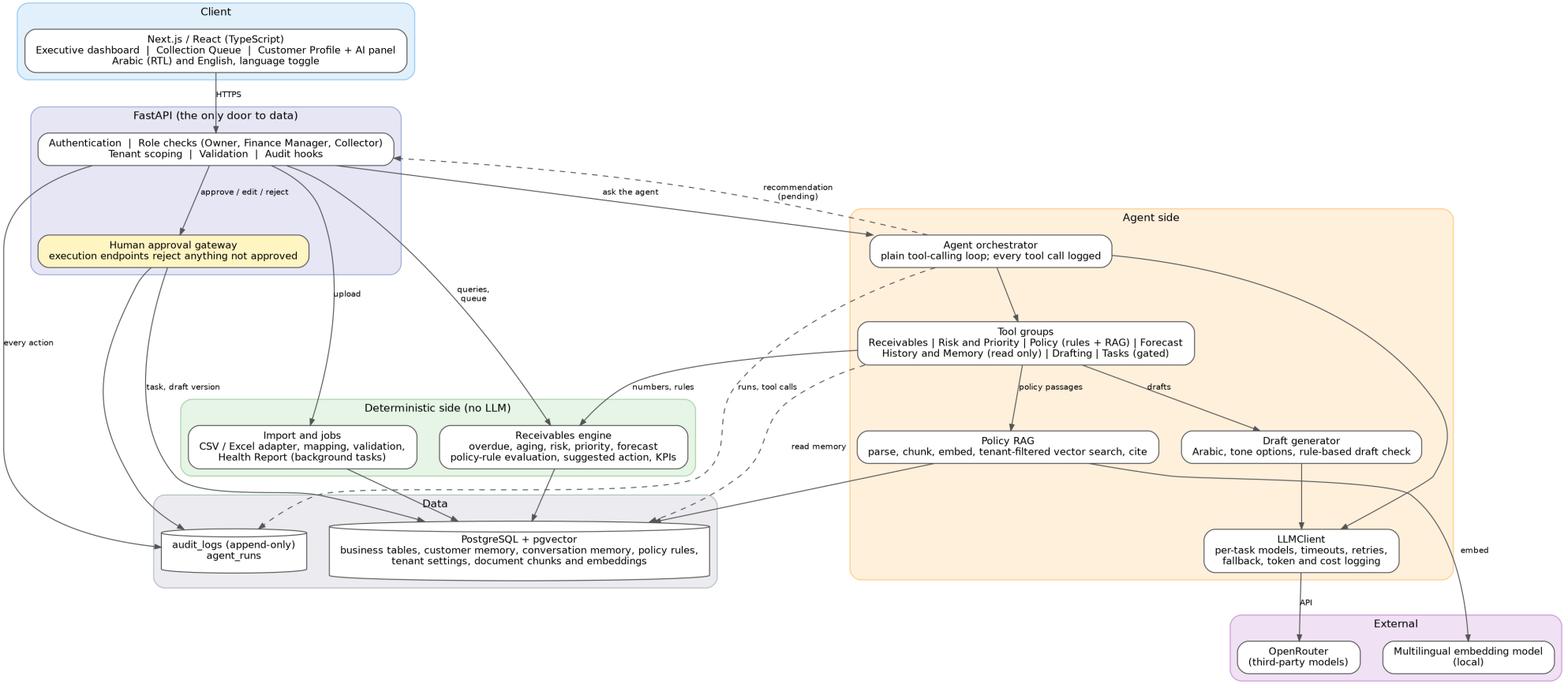
This part describes how Tahseel AI is built and why. The decisions behind it are in `docs/governance/Tahseel_AI_Decision_Log.md`; the trust design is in `docs/technical/Tahseel_AI_Technical_Strength.md`.

1. Design goal

Trust by construction. A finance team will not adopt a copilot it cannot verify, so the design goal is not "the smartest AI" but **the most checkable AI**:

- every number comes from tested code, never from the language model;
- every policy statement comes from a rule with a source or from a cited passage;
- every customer memory comes from a human-authored or rule-derived event;
- every consequential action waits for a human, and that is enforced at the API.

2. Logical architecture


Client
Next.js / React (TypeScript)
Executive dashboard | Collection Queue | Customer Profile + AI panel
Arabic (RTL) and English, language toggle
HTTPS
FastAPI (the only door to data)
Authentication | Role checks (Owner, Finance Manager, Collector)
Tenant scoping | Validation | Audit hooks
approve / edit / reject
Human approval gateway
execution endpoints reject anything not approved
ask the agent
recommendation
(pending)
Agent side
Agent orchestrator
plain tool-calling loop; every tool call logged
every action
upload
queries,
queue
Tool groups
Receivables | Risk and Priority | Policy (rules + RAG) | Forecast
History and Memory (read only) | Drafting | Tasks (gated)
task, draft version
numbers, rules
runs, tool calls
policy passages
drafts
Deterministic side (no LLM)
Import and jobs
CSV / Excel adapter, mapping, validation,
Health Report (background tasks)
Receivables engine
overdue, aging, risk, priority, forecast
policy-rule evaluation, suggested action, KPIs
read memory
Policy RAG
parse, chunk, embed, tenant-filtered vector search, cite
Draft generator
Arabic, tone options, rule-based draft check
Data
audit_logs (append-only)
agent_runs
PostgreSQL + pgvector
business tables, customer memory, conversation memory, policy rules,
tenant settings, document chunks and embeddings
LLMClient
per-task models, timeouts, retries,
fallback, token and cost logging
embed
API
External
OpenRouter
(third-party models)
Multilingual embedding model
(local)

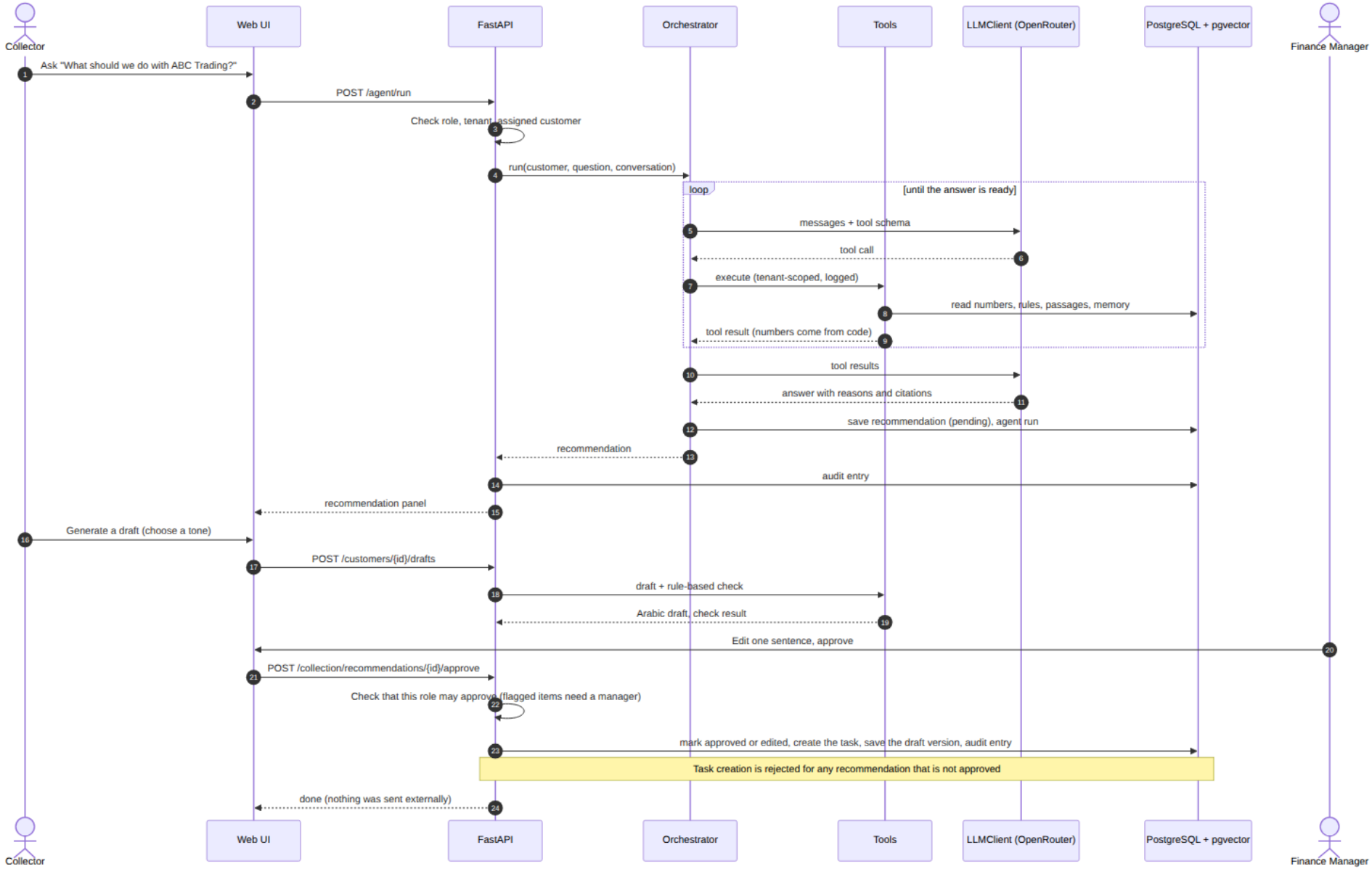
Layer	Responsibility	Technology
Client	Executive dashboard, Collection Queue, Customer Profile with the AI panel; fully Arabic (RTL) and English with a language toggle	Next.js / React, TypeScript
API	The only way to data: authentication, role checks, tenant scoping, validation, audit hooks, and the human approval gateway	FastAPI, Pydantic
Deterministic side	Overdue and aging, risk, priority, forecast, policy-rule evaluation, suggested action, KPIs, and the import with its Health Report. No LLM	Python, SQLAlchemy
Agent side	The orchestrator (a plain tool-calling loop), its tool groups, policy RAG, the draft generator, and the `LLMClient`	Python
External	Third-party models through OpenRouter; a local multilingual embedding model	OpenRouter API
Data	Business tables, both kinds of memory, policy rules, tenant settings, document chunks with embeddings, the append-only audit log, and agent runs	PostgreSQL with pgvector

3. What is a pipeline and what is an agent

Work	Mode	Why
Outstanding, days overdue, risk, priority, forecast for all customers	Batch pipeline, no LLM	Deterministic, cheap, unit-testable, reproducible
Policy rules evaluated per customer (thresholds, grace periods, escalation triggers)	Batch pipeline, no LLM	Hard thresholds must not depend on semantic search
Collection Queue with a rule-based suggested action	Batch pipeline	The same input gives the same output; no LLM cost per page load
Explaining a customer, answering open questions, investigating	Agent (the LLM selects and combines tools)	Open-ended work needs tool selection
Retrieving and citing policy passages, reading customer memory, drafting a message	Agent tools	Grounding, context, and a language task
Creating a task	Agent tool, blocked until a human approves	Consequential
Writing an audit entry	Always	Traceability; it never waits for approval
Recording a note, a call result, or a promise	A human, through the UI and the API	The agent never writes customer memory

The system is agentic where open-ended reasoning is needed and deterministic where correctness is needed. We do not call a fixed pipeline an agent. The queue shows a rule-based suggested action; the agent's recommendation can differ when memory or retrieved passages add context, and the interface shows both and says why.

4. How a request flows


Collector
Web UI
FastAPI
Orchestrator
Tools
LLMClient (OpenRouter)
PostgreSQL + pgvector
Finance Manager
Ask "What should we do with ABC Trading?"
POST /agent/run
Check role, tenant, assigned customer
run(customer, question, conversation)
loop
[until the answer is ready]
messages + tool schema
tool call
execute (tenant-scoped, logged)
read numbers, rules, passages, memory
tool result (numbers come from code)
tool results
answer with reasons and citations
save recommendation (pending), agent run
recommendation
audit entry
recommendation panel
Generate a draft (choose a tone)
POST /customers/{id}/drafts
draft + rule-based check
Arabic draft, check result
Edit one sentence, approve
POST /collection/recommendations/{id}/approve
Check that this role may approve (flagged items need a manager)
mark approved or edited, create the task, save the draft version, audit entry
Task creation is rejected for any recommendation that is not approved
done (nothing was sent externally)

The sequence shows the main use case: a Collector asks the agent about a customer, receives a recommendation with reasons and citations, generates an Arabic draft, and a Finance Manager approves it. Four properties hold at every step:

1. **Numbers come from tools.** The model quotes tool results; it does not calculate.
2. **Every tool call is tenant-scoped and logged.**
3. **The recommendation starts as pending.** Only a human decision changes it.
4. **Nothing is sent outside the system.** The approval creates a task and stores the draft; a person sends the message.

5. Trust boundaries and controls

Concern	Control
One company reading another's data	`tenant_id` on every table, one tenant-scoped data layer, the tenant filter inside every vector query, and a cross-tenant test
Too much access inside a company	Three roles enforced on the server; a Collector sees only assigned customers
The AI acting on its own	An approval gateway at the API: task creation is rejected for any recommendation that is not approved
Wrong or invented numbers	Deterministic engine; a post-check that every number in an answer appears in a tool result; an ablation measures the difference
Invented policy	Structured rules with sources, cited passages, and honest "no policy found" and "policy conflict" answers
Prompt injection	Retrieved text, notes, and uploaded documents are data, not instructions; no tool runs because a document says so
Untraceable decisions	An append-only audit log; the application role cannot update or delete it
Data sent to third-party models	Personal data kept minimal in prompts; synthetic data only in the demo; provider retention settings reviewed and providers restricted before any real data is used
Data in transit and at rest	TLS in transit; encryption at rest is a deployment setting; secrets stay out of the repository

6. Adaptability

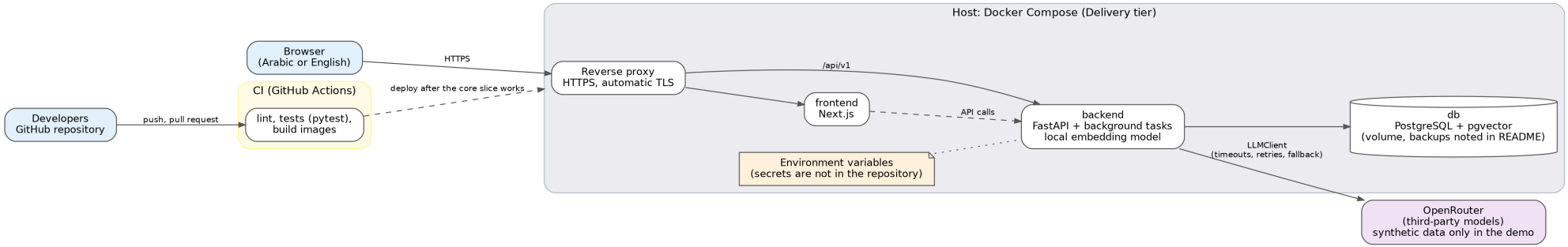
Sector differences are configuration, not code. The engine runs on a canonical schema; an adapter converts a source into it (the MVP builds the CSV and Excel adapter; Odoo and ETA adapters are roadmap). Risk weights, thresholds, grace periods, escalation triggers, the reminder cap, and tone defaults live in `tenant_settings`, and structured rules live in `policy_rules`. The details are in `docs/architecture/Tahseel_AI_Adaptability_Strategy.md`.

7. Key architecture decisions

ID	Decision	Reason
ADR-1	A deterministic engine, separate from the LLM	Correctness and testability
ADR-2	pgvector in the same PostgreSQL	One datastore; the tenant filter in the same query

ID	Decision	Reason
ADR-3	One orchestrator with grouped tools	Simpler to build, test, and defend
ADR-4	Approval enforced at the API	A user-interface bug cannot bypass it
ADR-5	An append-only audit log	Traceability
ADR-6	A multilingual embedding model	Policies and notes may be Arabic or mixed
ADR-7	Two-layer policy: rules evaluated by code, plus RAG for explanation	Thresholds must not depend on semantic search
ADR-8	A canonical schema with an adapter interface, and tenant configuration	Adaptability without code changes
ADR-9	LLM access through OpenRouter behind our own `LLMClient`	Switch models, compare cost, limit spend
ADR-10	A plain tool-calling loop	Fewer dependencies; every call can be logged and gated
ADR-11	Tenant isolation through one scoped data layer, proven by a test	Isolation that does not depend on each developer remembering a filter
ADR-12	Customer memory written only by humans and rules; conversation memory is context only	Keeps the trust thesis and the follow-up feature

8. Deployment


Host: Docker Compose (Delivery tier)
Browser
(Arabic or English)
HTTPS
Reverse proxy
HTTPS, automatic TLS
/api/v1
frontend
Next.js
API calls
backend
FastAPI + background tasks
local embedding model
db
PostgreSQL + pgvector
(volume, backups noted in README)
LLMClient
(timeouts, retries, fallback)
Environment variables
(secrets are not in the repository)
OpenRouter
(third-party models)
synthetic data only in the demo
CI (GitHub Actions)
deploy after the core slice works
Developers
GitHub repository
push, pull request
lint, tests (pytest),
build images

The deployment is part of the Delivery tier: it is scheduled after the core slice works, and a recorded demo is the fallback (Decision Log, D17 and D29). The basics are a Docker Compose setup with environment-based configuration, migrations, a seed script, a health-check endpoint, and CI that runs lint and tests. The public HTTPS deployment, structured logging, and the rest of the checklist come after.

9. Technology stack

Layer	Choice
Frontend	Next.js / React with TypeScript
Backend	Python, FastAPI, Pydantic, SQLAlchemy
Database	PostgreSQL with pgvector
AI	OpenRouter behind `LLMClient`; a plain tool-calling loop; a multilingual embedding model
Infrastructure	Docker Compose; GitHub Actions for CI
Tests	pytest for overdue, risk, priority, forecast, and tenant isolation; evaluation sets for agent, RAG, recommendations, and safety; the LLM-only versus tool-grounded ablation

10. Quality attributes

Attribute	How the design addresses it
Testability	The calculations are pure code with unit tests; the AI is evaluated with fixed sets and an ablation
Explainability	Every score lists its factors and points; every policy statement has a source
Security	The controls in section 5
Maintainability	Modules by responsibility; settings and rules as data; typed schemas
Internationalization	All strings in dictionaries from day one; logical CSS properties for RTL
Cost control	Page loads use no LLM; per-run token and cost logging; per-task model configuration

11. Limits of what we claim

- The numbers in the demo come from synthetic data. They are designed to show that the pipeline is correct, not that predictions are accurate on real distributors.
- No performance targets are claimed or measured yet.
- Numeric hallucination is reduced by construction and measured in the ablation; it is not eliminated.
- Cross-sector adaptability is a design property. It has not been validated on real sectors.

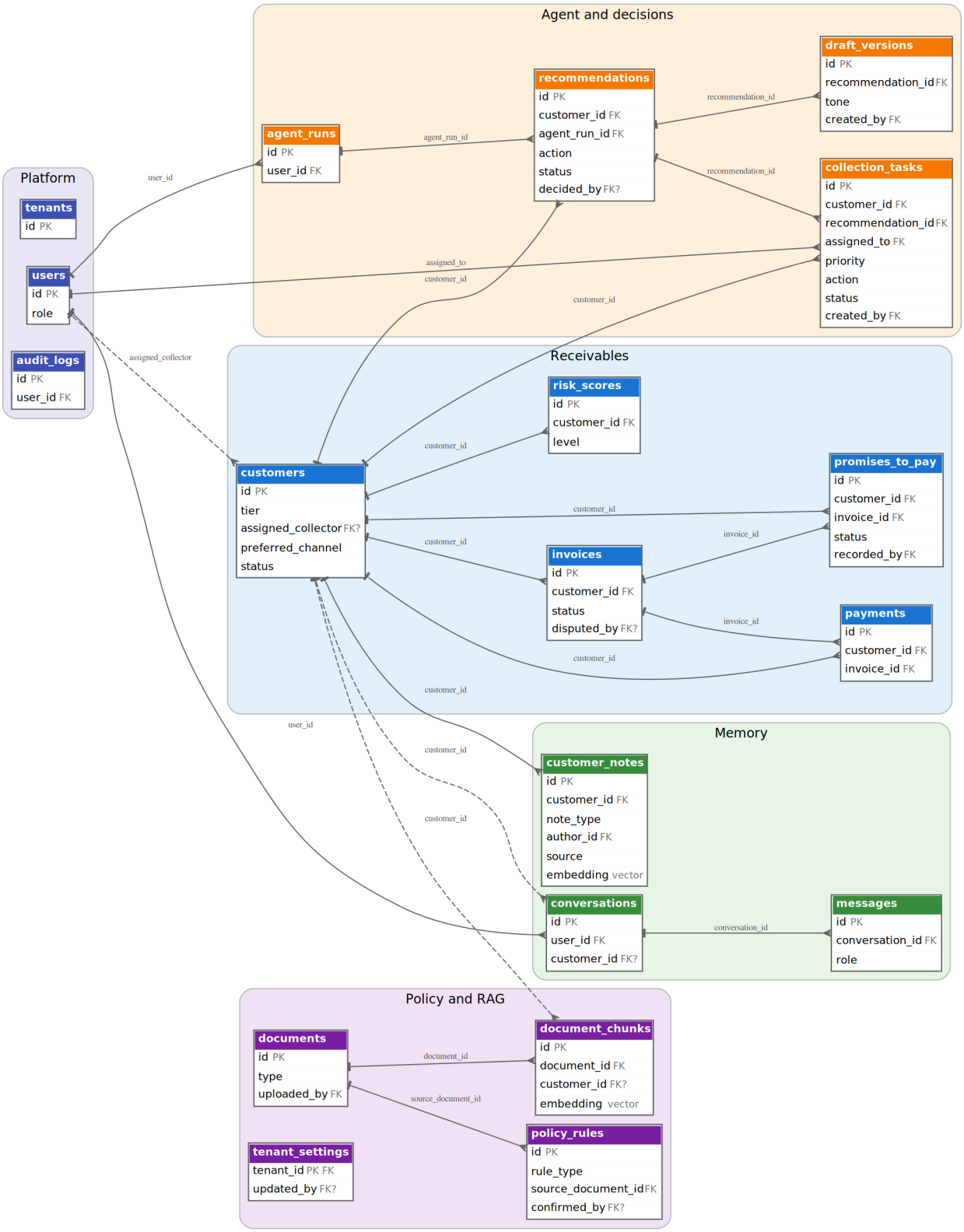
Database Design

This part describes the PostgreSQL schema of Tahseel AI. It is generated from the data model in the product specification (`docs/technical/Tahseel_AI_PRD.md`, section 19), so the two always agree.

1. Design principles

- **One database, tenant-scoped.** Every table except `tenants` carries `tenant_id`. All access goes through a tenant-scoped layer, and every vector query filters by tenant inside the query. A cross-tenant test proves it.
- **Payments are the source of truth.** An invoice's `paid_amount` is a cached total that the engine recomputes from `payments`. "Overdue" is computed from the due date, never stored.
- **Humans write customer memory.** Notes, call outcomes, and promises are written by people; promise status is derived by rule. The agent has no way to write them.
- **Append-only audit.** The application's database role has no update or delete permission on `audit_logs`. Deleting a memory entry is a soft delete (`deleted_at`) and is itself audited.
- **Vectors beside the data.** Embeddings live in pgvector columns (`customer_notes.embedding`, `document_chunks.embedding`), so the tenant filter and the relational filters run in one query.
- **Configuration, not code.** Weights, thresholds, and rules live in `tenant_settings` and `policy_rules`.

2. Entity-relationship diagram



Every table except `tenants` also has `tenant_id` referencing `tenants`; those edges are not drawn. The diagram shows keys and enumerated fields; the full field list is in section 4. A dashed line is an optional relationship. References from other tables to `users` (author, approver, creator) are listed in section 3 and not drawn.

3. Relationships

Child table	Field	References	Required
users	tenant_id	tenants	yes
customers	tenant_id	tenants	yes
customers	assigned_collector	users	no
invoices	tenant_id	tenants	yes
invoices	customer_id	customers	yes
invoices	disputed_by	users	no
payments	tenant_id	tenants	yes
payments	customer_id	customers	yes
payments	invoice_id	invoices	yes
promises_to_pay	tenant_id	tenants	yes
promises_to_pay	customer_id	customers	yes
promises_to_pay	invoice_id	invoices	yes
promises_to_pay	recorded_by	users	yes
customer_notes	tenant_id	tenants	yes
customer_notes	customer_id	customers	yes
customer_notes	author_id	users	yes
recommendations	tenant_id	tenants	yes
recommendations	customer_id	customers	yes
recommendations	agent_run_id	agent_runs	yes
recommendations	decided_by	users	no
draft_versions	tenant_id	tenants	yes
draft_versions	recommendation_id	recommendations	yes
draft_versions	created_by	users	yes
collection_tasks	tenant_id	tenants	yes
collection_tasks	customer_id	customers	yes
collection_tasks	recommendation_id	recommendations	yes
collection_tasks	assigned_to	users	yes
collection_tasks	created_by	users	yes
conversations	tenant_id	tenants	yes
conversations	user_id	users	yes
conversations	customer_id	customers	no
messages	tenant_id	tenants	yes
messages	conversation_id	conversations	yes
documents	tenant_id	tenants	yes
documents	uploaded_by	users	yes

Child table	Field	References	Required
document_chunks	tenant_id	tenants	yes
document_chunks	document_id	documents	yes
document_chunks	customer_id	customers	no
policy_rules	tenant_id	tenants	yes
policy_rules	source_document_id	documents	yes
policy_rules	confirmed_by	users	no
tenant_settings	tenant_id	tenants	yes
tenant_settings	updated_by	users	no
risk_scores	tenant_id	tenants	yes
risk_scores	customer_id	customers	yes
agent_runs	tenant_id	tenants	yes
agent_runs	user_id	users	yes
audit_logs	tenant_id	tenants	yes
audit_logs	user_id	users	yes

4. Data dictionary

Types are **proposed** and are finalized in the first migration. `uuid` keys and `timestamptz` for moments are the working choices. `vector(dim)` takes its size from the embedding model chosen after the Arabic retrieval test (Decision Log, D16). Fields with a fixed set of values are stored as text with a CHECK constraint.

tenants

One customer company on the platform. Every other table points to it.

Field	Type	Key	Notes
id	uuid	PK	
name	text		
industry	text		
created_at	timestamptz		

users

People who sign in. Each has one of three roles.

Field	Type	Key	Notes
id	uuid	PK	
tenant_id	uuid	FK	
name	text		
email	text		

Field	Type	Key	Notes
password_hash	text		
role	text		values: OWNER / FINANCE_MANAGER / COLLECTOR
created_at	timestamptz		

customers

The distributor's credit customers, with tier, sales rep, assigned collector, and contact preferences.

Field	Type	Key	Notes
id	uuid	PK	
tenant_id	uuid	FK	
name	text		
phone	text		
email	text		
credit_limit	numeric(14,2)		stored as data; recommending limits is out of scope
payment_terms	text		
tier	text		values: strategic / regular
sales_rep	text		
assigned_collector	uuid	FK?	user id; makes "assigned only" enforceable
preferred_channel	text		values: call / whatsapp / email; set by a human
contact_person	text		set by a human
status	text		values: active / inactive
created_at	timestamptz		

invoices

Invoices issued to customers. Overdue is computed from the due date and never stored.

Field	Type	Key	Notes
id	uuid	PK	
tenant_id	uuid	FK	
customer_id	uuid	FK	
invoice_number	text		
invoice_date	date		
due_date	date		
amount	numeric(14,2)		
paid_amount	numeric(14,2)		cache, recomputed from payments
status	text		values: open / paid / disputed
dispute_reason	text		
disputed_by	uuid	FK?	

Field	Type	Key	Notes
disputed_at	timestamptz		
created_at	timestamptz		

payments

Payments received. The source of truth for what has been paid.

Field	Type	Key	Notes
id	uuid	PK	
tenant_id	uuid	FK	
customer_id	uuid	FK	
invoice_id	uuid	FK	
amount	numeric(14,2)		
payment_date	date		
method	text		
created_at	timestamptz		

promises_to_pay

Promises recorded by a person. Their status is derived by rule from payments.

Field	Type	Key	Notes
id	uuid	PK	
tenant_id	uuid	FK	
customer_id	uuid	FK	
invoice_id	uuid	FK	
promised_amount	numeric(14,2)		
promise_date	date		
status	text		values: pending / kept / broken
recorded_by	uuid	FK	
created_at	timestamptz		
resolved_at	timestamptz		

customer_notes

Customer memory: notes, sales rep notes, and call outcomes, written by humans, with an embedding for recall.

Field	Type	Key	Notes
id	uuid	PK	
tenant_id	uuid	FK	
customer_id	uuid	FK	

Field	Type	Key	Notes
`note_type`	text		values: collector / sales_rep / call_outcome
`text`	text		
`embedding`	vector(dim)		vector; D16
`author_id`	uuid	FK	
`source`	text		values: ui / import
`created_at`	timestamptz		
`deleted_at`	timestamptz		soft delete; see the audit log

recommendations

What the agent recommended and what a human decided (approved, edited, rejected).

Field	Type	Key	Notes
`id`	uuid	PK	
`tenant_id`	uuid	FK	
`customer_id`	uuid	FK	
`agent_run_id`	uuid	FK	
`action`	text		values: CALL / WHATSAPP / EMAIL / WAIT / ESCALATE_TO_MANAGER / TALK_TO_SALES_REP
`reasons`	jsonb		list of reasons
`citations`	jsonb		triggered rules and retrieved passages
`requires_manager_review`	boolean		
`status`	text		values: pending / approved / edited / rejected
`decided_by`	uuid	FK?	
`decided_at`	timestamptz		
`decision_reason`	text		
`created_at`	timestamptz		

draft_versions

The original and edited text of each message draft, with the reason and the draft-check result.

Field	Type	Key	Notes
`id`	uuid	PK	
`tenant_id`	uuid	FK	
`recommendation_id`	uuid	FK	
`tone`	text		values: formal / friendly / firm
`original_text`	text		
`edited_text`	text		
`edit_reason`	text		

Field	Type	Key	Notes
`check_result`	jsonb		output of the rule-based draft check
`created_by`	uuid	FK	
`created_at`	timestamptz		

collection_tasks

Follow-up tasks created from approved recommendations.

Field	Type	Key	Notes
`id`	uuid	PK	
`tenant_id`	uuid	FK	
`customer_id`	uuid	FK	
`recommendation_id`	uuid	FK	
`assigned_to`	uuid	FK	
`priority`	text		values: HIGH / MEDIUM / LOW
`action`	text		values: CALL / WHATSAPP / EMAIL / WAIT / ESCALATE_TO_MANAGER / TALK_TO_SALES_REP
`due_at`	timestamptz		
`status`	text		values: pending / done / cancelled
`notes`	text		
`created_by`	uuid	FK	
`created_at`	timestamptz		

conversations

Chat threads between a user and the agent (conversation memory).

Field	Type	Key	Notes
`id`	uuid	PK	
`tenant_id`	uuid	FK	
`user_id`	uuid	FK	
`customer_id`	uuid	FK?	optional
`created_at`	timestamptz		

messages

Turns of a thread; an LLM-written summary is marked and never treated as fact.

Field	Type	Key	Notes
`id`	uuid	PK	
`tenant_id`	uuid	FK	
`conversation_id`	uuid	FK	

Field	Type	Key	Notes
role	text		values: user / assistant / tool
content	text		
tool_calls	jsonb		
is_summary	boolean		true for an LLM-written running summary, labeled AI-generated
created_at	timestamptz		

documents

Uploaded policy, contract, template, and terms documents.

Field	Type	Key	Notes
id	uuid	PK	
tenant_id	uuid	FK	
name	text		
type	text		values: credit_policy / collection_policy / payment_terms / template / contract / other
storage_path	text		
uploaded_by	uuid	FK	
effective_date	date		
created_at	timestamptz		

document_chunks

Searchable passages of documents with an embedding, for cited retrieval.

Field	Type	Key	Notes
id	uuid	PK	
tenant_id	uuid	FK	
document_id	uuid	FK	
customer_id	uuid	FK?	optional
policy_type	text		
section	text		
page	integer		
effective_date	date		
text	text		
embedding	vector(dim)		vector
created_at	timestamptz		

policy_rules

Structured rules evaluated by code, each with its source document and section.

Field	Type	Key	Notes
`id`	uuid	PK	
`tenant_id`	uuid	FK	
`name`	text		
`rule_type`	text		values: escalation / grace_period / limit / review_required
`condition`	jsonb		
`outcome`	jsonb		for example requires_manager_review, or a suggested action
`source_document_id`	uuid	FK	
`source_section`	text		
`confirmed_by`	uuid	FK?	
`confirmed_at`	timestamptz		
`active`	boolean		

tenant_settings

Per-tenant configuration: weights, caps, grace periods, triggers, reminder cap, tone defaults.

Field	Type	Key	Notes
`tenant_id`	uuid	PK FK	
`settings`	jsonb		validated by a typed schema
`updated_by`	uuid	FK?	
`updated_at`	timestamptz		

risk_scores

The latest and past risk scores with the points of every factor.

Field	Type	Key	Notes
`id`	uuid	PK	
`tenant_id`	uuid	FK	
`customer_id`	uuid	FK	
`score`	smallint		
`level`	text		values: LOW / MEDIUM / HIGH / CRITICAL
`factors`	jsonb		factor, points, detail
`calculated_at`	timestamptz		

agent_runs

One row per agent run, with model, tools used, tokens, cost, and latency.

Field	Type	Key	Notes
`id`	uuid	PK	

Field	Type	Key	Notes
`tenant_id`	uuid	FK	
`user_id`	uuid	FK	
`run_type`	text		
`input`	jsonb		
`output`	jsonb		
`tools_used`	jsonb		
`status`	text		
`model`	text		
`prompt_tokens`	integer		
`completion_tokens`	integer		
`cost_estimate`	numeric(10,4)		
`latency_ms`	integer		
`created_at`	timestamptz		

audit_logs

Append-only record of every recommendation, approval, edit, rejection, tool call, and memory change.

Field	Type	Key	Notes
`id`	uuid	PK	
`tenant_id`	uuid	FK	
`user_id`	uuid	FK	
`action`	text		
`entity_type`	text		
`entity_id`	text		
`before`	jsonb		
`after`	jsonb		
`created_at`	timestamptz		

5. Indexes and constraints (proposed)

Where	What	Why
`invoices`	unique (`tenant_id`, `invoice_number`)	The same invoice number cannot be imported twice by one tenant
`invoices`	index (`tenant_id`, `customer_id`, `due_date`)	Fast overdue queries per customer
`payments`	index (`tenant_id`, `invoice_id`)	Recompute `paid_amount` quickly
`customer_notes`, `document_chunks`	vector index on `embedding` plus btree on `tenant_id`	Similarity search that always filters by tenant

Where	What	Why
`audit_logs`	no UPDATE or DELETE grant for the application role	Append-only
`tenant_settings`	the risk weights total 100 (checked by the typed settings schema)	The score arithmetic stays valid
enumerated fields	CHECK constraint on the allowed values	Bad values are rejected by the database

6. Open points

- The embedding dimension and the model are not chosen yet (D16).
- Invoice line items are not modeled; the MVP works at invoice level (see the Adaptability Strategy).
- Types, indexes, and constraints above are a design proposal and will be fixed when the first migration is written.

UI/UX Design

This part shows how Tahseel AI looks and behaves. The wireframes are low fidelity: they fix the layout, the content, and the behavior, not the final visual style. Every screen exists in English and in Arabic (right to left). All figures on the wireframes are **sample data for layout only; they are not results**.

1. Design principles

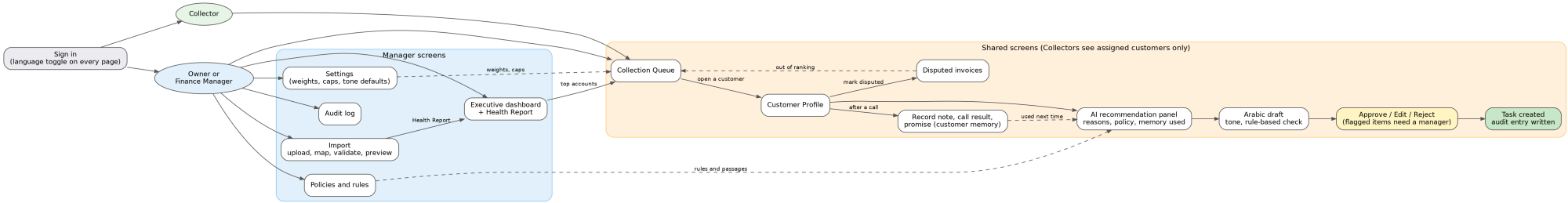
Principle	What it means on screen
Show the reason, not only the answer	Every score lists its factors and points; every recommendation lists why, which policy rule or passage applies, and which memory was used
The human decides	Approve, Edit, and Reject are always visible; the screen says plainly that Tahseel never sends the message
Honest "I cannot" states	"No relevant policy found", "policies conflict", "invoice disputed", and "reminder cap reached" are designed states, not error pages
One list to start the day	The Collection Queue is the Collector's home; the executive dashboard is the manager's
Fully bilingual	Arabic (right to left) and English with a language toggle on every page; no text is hard-coded
Right information for the role	A Collector sees assigned customers; managers see everything, settings, and the audit log

2. Users and their main tasks

Role	Main tasks	Home screen
Collector	See who to chase today; open a customer; ask the agent; generate and submit a draft; record a call result and a promise; mark an invoice as disputed	Collection Queue
Finance Manager	Review flagged recommendations; approve or reject drafts; confirm policy rules; set weights and caps; import data	Executive dashboard
Owner	See the summary and cash expectations; manage users and roles	Executive dashboard

The personas and their needs are in the Requirements part. Every need there is a hypothesis until interviews happen.

3. Navigation and task flow


Collector
Sign in
(language toggle on every page)
Owner or
Finance Manager
Manager screens
Settings
(weights, caps, tone defaults)
Audit log
Import
upload, map, validate, preview
Policies and rules
Executive dashboard
+ Health Report
weights, caps
Health Report
top accounts
Shared screens (Collectors see assigned customers only)
Collection Queue
out of ranking
open a customer
Disputed invoices
mark disputed
Customer Profile
after a call
Record note, call result,
promise (customer memory)
used next time
AI recommendation panel
reasons, policy, memory used
Arabic draft
tone, rule-based check
Approve / Edit / Reject
(flagged items need a manager)
Task created
audit entry written
rules and passages

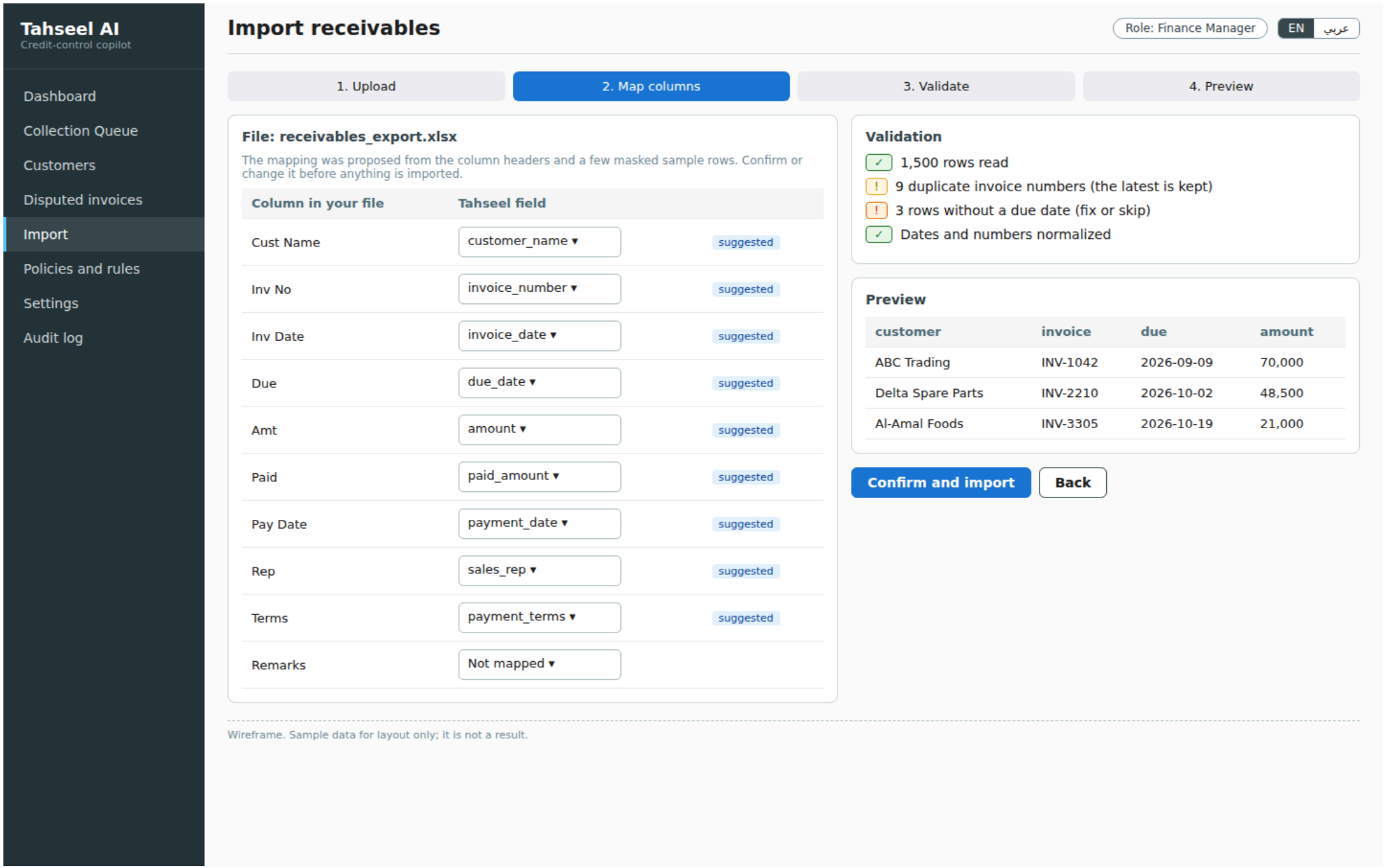
The main path follows the demo flow in the Final Concept: import, Health Report, Collection Queue, Customer Profile, the AI panel, the Arabic draft, the decision, and the task with its audit entry. Recording a call result feeds the memory that the AI panel uses the next time.

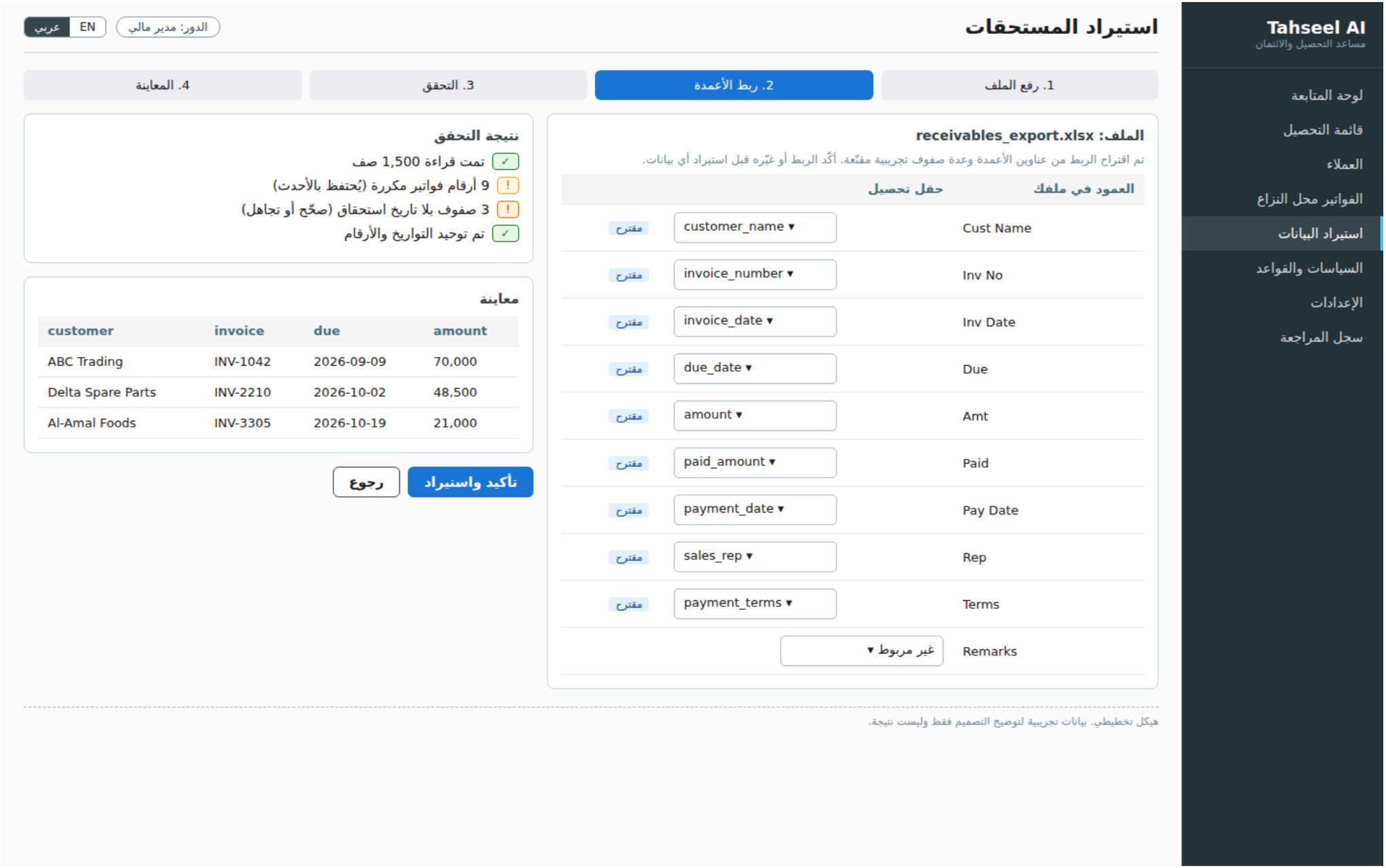
4. Screens

Each screen is shown in English and in Arabic. The Arabic version mirrors the layout (the navigation moves to the right) and keeps Western digits.

4.1 Import receivables

Capability C1 and the start of the demo. The user uploads a CSV or Excel file, the column mapping is proposed from the headers and a few masked sample rows, and the user confirms it. Validation lists problems before anything is imported, and a preview shows how the rows will look.


Tahseel AI
Credit-control copilot
Dashboard
Collection Queue
Customers
Disputed invoices
Import
Policies and rules
Settings
Audit log
Import receivables
Role: Finance Manager
EN
عربي
1. Upload
2. Map columns
3. Validate
4. Preview
File: receivables_export.xlsx
The mapping was proposed from the column headers and a few masked sample rows. Confirm or change it before anything is imported.
Column in your file
Tahseel field
Cust Name
customer_name ▾
suggested
Inv No
invoice_number ▾
suggested
Inv Date
invoice_date ▾
suggested
Due
due_date ▾
suggested
Amt
amount ▾
suggested
Paid
paid_amount ▾
suggested
Pay Date
payment_date ▾
suggested
Rep
sales_rep ▾
suggested
Terms
payment_terms ▾
suggested
Remarks
Not mapped ▾
Validation
1,500 rows read
9 duplicate invoice numbers (the latest is kept)
3 rows without a due date (fix or skip)
Dates and numbers normalized
Preview
customer
invoice
due
amount
ABC Trading
INV-1042
2026-09-09
70,000
Delta Spare Parts
INV-2210
2026-10-02
48,500
Al-Amal Foods
INV-3305
2026-10-19
21,000
Confirm and import
Back
Wireframe. Sample data for layout only; it is not a result.


Tahseel AI
مساعد التحصيل والائتمان
لوحة المتابعة
قائمة التحصيل
العملاء
الفواتير محل النزاع
استيراد البيانات
السياسات والقواعد
الإعدادات
سجل المراجعة
استيراد المستحقات
الدور: مدير مالي
EN
عربي
1. رفع الملف
2. ربط الأعمدة
3. التحقق
4. المعاينة
الملف: receivables_export.xlsx
تم اقتراح الربط من عناوين الأعمدة وعدة صفوف تجريبية مقنّعة. أكّد الربط أو غيّره قبل استيراد أي بيانات.
العمود في ملفك
حقل تحصيل
Cust Name
customer_name
مقترح
Inv No
invoice_number
مقترح
Inv Date
invoice_date
مقترح
Due
due_date
مقترح
Amt
amount
مقترح
Paid
paid_amount
مقترح
Pay Date
payment_date
مقترح
Rep
sales_rep
مقترح
Terms
payment_terms
مقترح
Remarks
غير مربوط
نتيجة التحقق
تمت قراءة 1,500 صف
9 أرقام فواتير مكررة (يُحتفظ بالأحدث)
3 صفوف بلا تاريخ استحقاق (صحّح أو تجاهل)
تم توحيد التواريخ والأرقام
معاينة
customer
invoice
due
amount
ABC Trading
INV-1042
2026-09-09
70,000
Delta Spare Parts
INV-2210
2026-10-02
48,500
Al-Amal Foods
INV-3305
2026-10-19
21,000
تأكيد واستيراد
رجوع
هيكل تخطيطي. بيانات تجريبية لتوضيح التصميم فقط وليست نتيجة.

4.2 Executive dashboard

For the Owner and the Finance Manager (Core, with the forecast and Health Report in the Should tier). It opens with the Receivables Health Report banner after the first upload. The cards show total receivables, the overdue amount, the high-risk exposure, and the average days beyond terms with DSO as a secondary figure. The expected cash is labeled "not guaranteed" and shows a range and a confidence label.

Tahseel AI
Credit-control copilot

- Dashboard
- Collection Queue
- Customers
- Disputed invoices
- Import
- Policies and rules
- Settings
- Audit log

Executive dashboard

Role: Finance Manager | EN | عربي

Receivables Health Report is ready · Generated from your first upload. Open the report

Total receivables
EGP 10.0M
Outstanding, all customers

Overdue
EGP 2.4M
24% of receivables

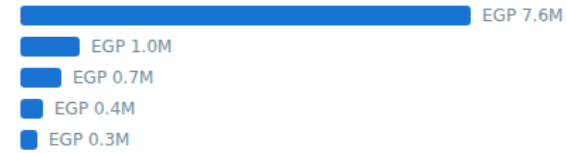
High-risk exposure
EGP 0.9M
Customers rated High or Critical

Average days beyond terms
38
DSO 71 days (secondary)

Expected cash (not guaranteed)

	Expected	Range	Confidence
Next 7 days	EGP 1.1M	0.8 – 1.3	Medium
Next 30 days	EGP 3.9M	3.2 – 4.4	Medium

Aging of receivables



Top accounts to chase today

Customer	Overdue	Risk	Priority	Suggested action
ABC Trading	EGP 310K	High · 70	100	Call
Delta Spare Parts	EGP 190K	Medium · 58	38	WhatsApp draft
Nile Building Supplies	EGP 150K	High · 66	23	Call
Al-Amal Foods	EGP 85K	Medium · 41	12	Wait
Cairo Medical Supplies Strategic	EGP 120K	Medium · 34	9	Talk to the sales rep first

Overdue trend

line chart

Promise-to-pay fulfillment
64%

Wireframe. Sample data for layout only; it is not a result.

لوحة المتابعة التنفيذية

تقرير صحة المستحقات جاهز · تم إنشاؤه من أول ملف رفعته. افتح التقرير

إجمالي المستحقات
10.0 مليون ج.م
الرصيد المستحق لكل العملاء

متأخر
2.4 مليون ج.م
24% من المستحقات

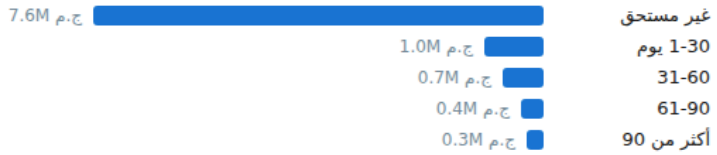
التعرض عالي المخاطرة
0.9 مليون ج.م
عملاء بدرجة مرتفع أو حرج

متوسط أيام التأخير عن الآجل
38
متوسط أيام التحصيل 71 يوماً (ثانوي)

النقدية المتوقعة (غير مضمونة)

	المتوقع	النطاق	الثقة
خلال 7 أيام	1.1 مليون	1.3 – 0.8	متوسطة
خلال 30 يوماً	3.9 مليون	4.4 – 3.2	متوسطة

تقادم المستحقات



أهم الحسابات للمتابعة اليوم

العميل	المتأخر	المخاطرة	الأولوية	الإجراء المقترح
شركة ABC للتجارة	310 ألف ج.م	مرتفع · 70	100	اتصال
دلتا لقطع الغيار	190 ألف ج.م	متوسط · 58	38	مسودة واتساب
النيل لمواد البناء	150 ألف ج.م	مرتفع · 66	23	اتصال
الأمل للأغذية	85 ألف ج.م	متوسط · 41	12	انتظار
القاهرة للمستلزمات الطبية استراتيجي	120 ألف ج.م	متوسط · 34	9	كلّم مندوب المبيعات أولاً

اتجاه المتأخرات

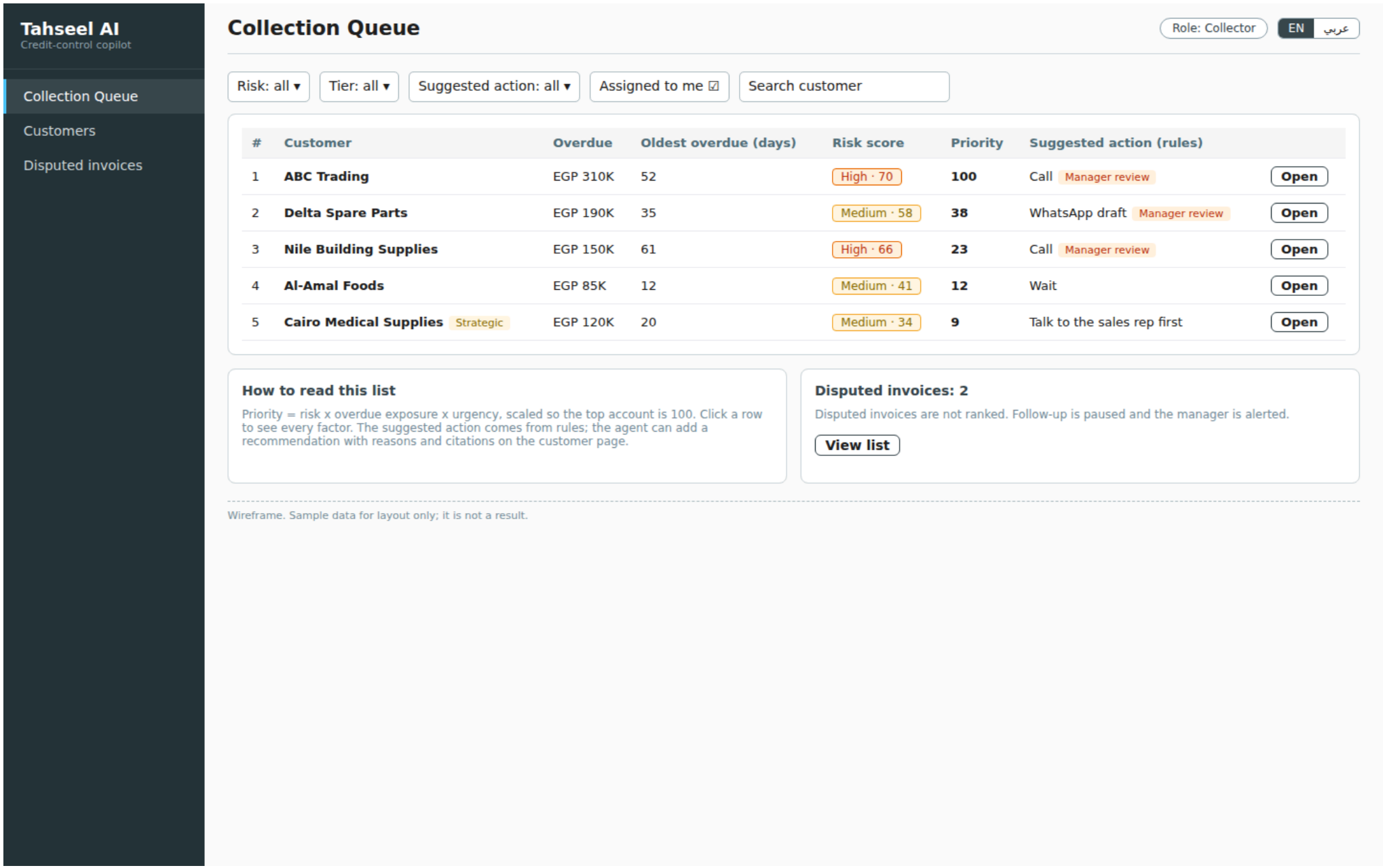
رسم خطي

الوفاء بوعود الدفع
64%

هيكل تخطيطي. بيانات تجريبية لتوضيح التصميم فقط وليست نتيجة.

4.3 Collection Queue

The Collector's home screen. Customers are ranked by priority. Each row shows the overdue amount, the oldest overdue invoice, the risk score as a labeled badge, the priority, and the **suggested action computed by rules**. A tag marks items that need manager review and strategic accounts, whose suggested action is "talk to the sales rep first". Disputed invoices are listed separately and are not ranked.


Tahseel AI
Credit-control copilot
Collection Queue
Customers
Disputed invoices
Collection Queue
Role: Collector
EN
عربي
Risk: all ▾
Tier: all ▾
Suggested action: all ▾
Assigned to me ☑
Search customer
Customer Overdue Oldest overdue (days) Risk score Priority Suggested action (rules)
1 ABC Trading EGP 310K 52 High · 70 100 Call Manager review Open
2 Delta Spare Parts EGP 190K 35 Medium · 58 38 WhatsApp draft Manager review Open
3 Nile Building Supplies EGP 150K 61 High · 66 23 Call Manager review Open
4 Al-Amal Foods EGP 85K 12 Medium · 41 12 Wait Open
5 Cairo Medical Supplies Strategic EGP 120K 20 Medium · 34 9 Talk to the sales rep first Open
How to read this list
Priority = risk x overdue exposure x urgency, scaled so the top account is 100. Click a row to see every factor. The suggested action comes from rules; the agent can add a recommendation with reasons and citations on the customer page.
Disputed invoices: 2
Disputed invoices are not ranked. Follow-up is paused and the manager is alerted.
View list
Wireframe. Sample data for layout only; it is not a result.

قائمة التحصيل

عربي | EN — الدور: محصّل

المخاطرة: الكل ▾ — الفئة: الكل ▾ — الإجراء: الكل ▾ — ☑ المسندة لي — ابحث عن عميل

#	العميل	المتأخر	أقدم تأخير (أيام)	درجة المخاطرة	الأولوية	الإجراء المقترح (بالقواعد)	
1	**شركة ABC للتجارة**	310 ألف ج.م	52	مرتفع · 70	**100**	اتصال — مراجعة المدير	فتح
2	**دلتا لقطع الغيار**	190 ألف ج.م	35	متوسط · 58	**38**	مسودة واتساب — مراجعة المدير	فتح
3	**النيل لمواد البناء**	150 ألف ج.م	61	مرتفع · 66	**23**	اتصال — مراجعة المدير	فتح
4	**الأمل للأغذية**	85 ألف ج.م	12	متوسط · 41	**12**	انتظار	فتح
5	**القاهرة للمستلزمات الطبية** استراتيجي	120 ألف ج.م	20	متوسط · 34	**9**	كلّم مندوب المبيعات أولاً	فتح

كيف تقرأ هذه القائمة

الأولوية = المخاطرة × المتأخر × الاستعجال، ومقيسة بحيث يكون أعلى حساب 100. اضغط على الصف لترى كل العوامل. الإجراء المقترح يأتي من القواعد، وقد يضيف الوكيل توصية بأسبابها ومصادرها في صفحة العميل.

فواتير محل نزاع: 2

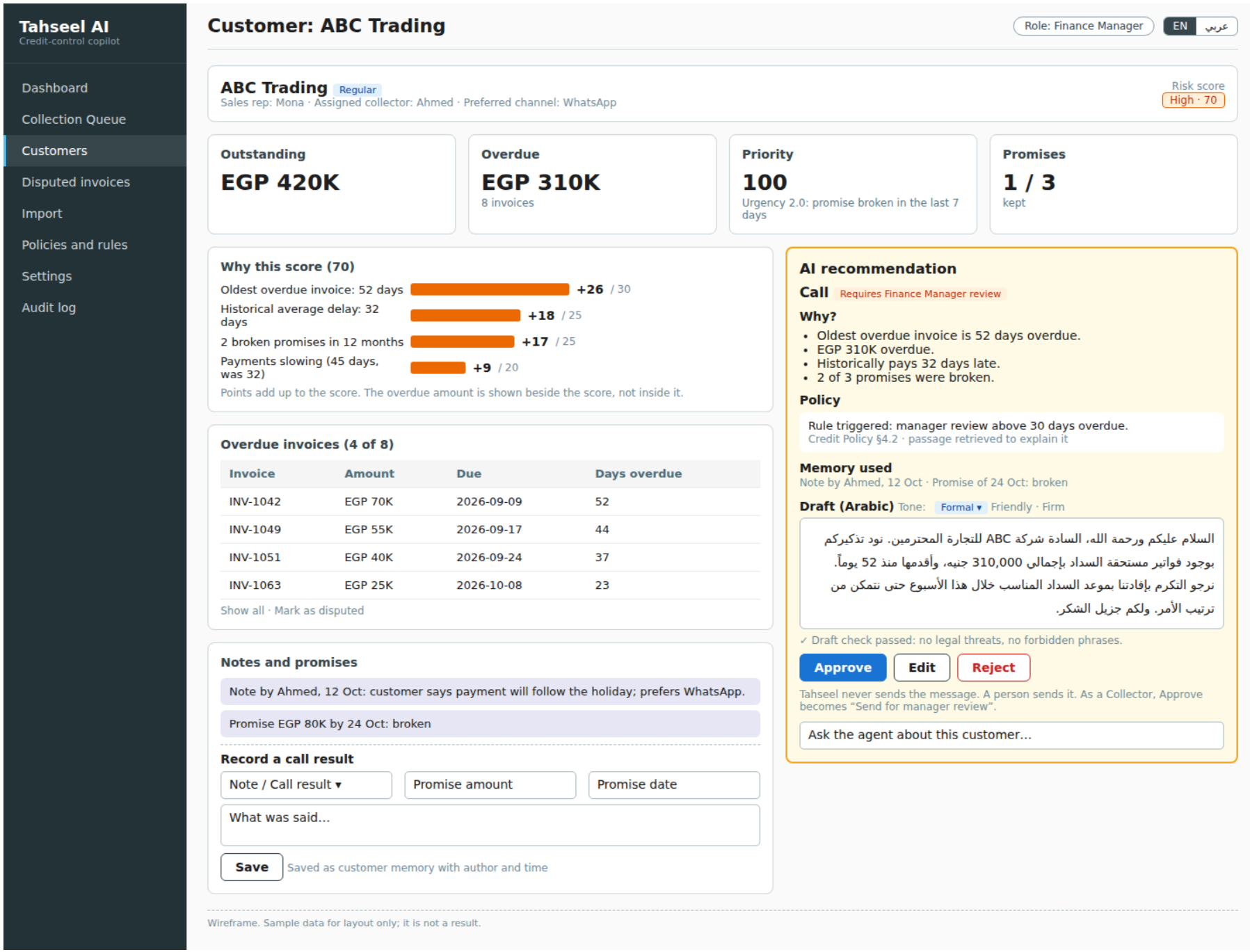
الفواتير محل النزاع لا تدخل الترتيب. المتابعة متوقفة والمدير تم تنبيهه.

عرض القائمة

هيكل تخطيطي. بيانات تجريبية لتوضيح التصميم فقط وليست نتيجة.

4.4 Customer Profile with the AI panel

The densest screen. On the left side (right in Arabic): the customer's summary, the score with every factor and its points, the overdue invoices, the notes and promises, and the form to record a call result. On the other side: the **AI recommendation panel** with the recommended action, the reasons, the policy rule and the passage that explains it, the memory that was used, the Arabic draft with tone options and the result of the rule-based draft check, and the Approve, Edit, and Reject buttons. A Collector sees "Send for manager review" in place of Approve for flagged items.


Tahseel AI
Credit-control copilot
Dashboard
Collection Queue
Customers
Disputed invoices
Import
Policies and rules
Settings
Audit log
Customer: ABC Trading
Role: Finance Manager
EN
عربي
ABC Trading
Regular
Sales rep: Mona · Assigned collector: Ahmed · Preferred channel: WhatsApp
Risk score
High · 70
Outstanding
EGP 420K
Overdue
EGP 310K
8 invoices
Priority
100
Urgency 2.0: promise broken in the last 7 days
Promises
1 / 3
kept
Why this score (70)
Oldest overdue invoice: 52 days +26 / 30
Historical average delay: 32 days +18 / 25
2 broken promises in 12 months +17 / 25
Payments slowing (45 days, was 32) +9 / 20
Points add up to the score. The overdue amount is shown beside the score, not inside it.
Overdue invoices (4 of 8)
Invoice
Amount
Due
Days overdue
INV-1042
EGP 70K
2026-09-09
52
INV-1049
EGP 55K
2026-09-17
44
INV-1051
EGP 40K
2026-09-24
37
INV-1063
EGP 25K
2026-10-08
23
Show all · Mark as disputed
Notes and promises
Note by Ahmed, 12 Oct: customer says payment will follow the holiday; prefers WhatsApp.
Promise EGP 80K by 24 Oct: broken
Record a call result
Note / Call result ▾
Promise amount
Promise date
What was said...
Save
Saved as customer memory with author and time
AI recommendation
Call
Requires Finance Manager review
Why?
Oldest overdue invoice is 52 days overdue.
EGP 310K overdue.
Historically pays 32 days late.
2 of 3 promises were broken.
Policy
Rule triggered: manager review above 30 days overdue.
Credit Policy §4.2 · passage retrieved to explain it
Memory used
Note by Ahmed, 12 Oct · Promise of 24 Oct: broken
Draft (Arabic) Tone: Formal ▾ Friendly · Firm
السلام عليكم ورحمة الله، السادة شركة ABC للتجارة المحترمين. نود تذكيركم بوجود فواتير مستحقة السداد بإجمالي 310,000 جنيه، وأقدمها منذ 52 يوماً.
نرجو التكرم بإفادتنا بموعد السداد المناسب خلال هذا الأسبوع حتى نتمكن من ترتيب الأمر. ولكم جزيل الشكر.
✓ Draft check passed: no legal threats, no forbidden phrases.
Approve
Edit
Reject
Tahseel never sends the message. A person sends it. As a Collector, Approve becomes “Send for manager review”.
Ask the agent about this customer...
Wireframe. Sample data for layout only; it is not a result.

العميل: شركة ABC للتجارة

عربي | EN — الدور: مدير مالي

شركة ABC للتجارة عادي

مندوبة المبيعات: منى · المحصّل المسؤول: أحمد · القناة المفضلة: واتساب

درجة المخاطرة: مرتفع · 70

الرصيد المستحق	المتأخر	الأولوية	الوعود
420 ألف ج.م	310 ألف ج.م 8 فواتير	100 الاستعجال 2.0: وعد انكسر خلال آخر 7 أيام	3 / 1 تم الوفاء بها

سبب الدرجة (70)

أقدم فاتورة متأخرة: 52 يوماً	+26	30 /
متوسط التأخير التاريخي: 32 يوماً	+18	25 /
وعدان مكسوران خلال 12 شهراً	+17	25 /
الدفع بتباطأ (45 يوماً بعد 32)	+9	20 /

النقاط تجمع إلى الدرجة. المبلغ المتأخر يظهر بجانب الدرجة وليس داخلها.

الفواتير المتأخرة (4 من 8)

الفاتورة	المبلغ	الاستحقاق	أيام التأخير
INV-1042	70 ألف ج.م	2026-09-09	52
INV-1049	55 ألف ج.م	2026-09-17	44
INV-1051	40 ألف ج.م	2026-09-24	37
INV-1063	25 ألف ج.م	2026-10-08	23

عرض الكل · تعليم كنزاع

الملاحظات والوعود

ملاحظة من أحمد، 12 أكتوبر: العميل يفيد بأن السداد سيكون بعد الإجازة؛ يفضّل واتساب.

وعد 80 ألف ج.م حتى 24 أكتوبر: مكسور

سجّل نتيجة مكالمة

ملاحظة / نتيجة مكالمة ▾ | مبلغ الوعد | تاريخ الوعد

ماذا دار في المكالمة...

حفظ — تُحفظ كذاكرة للعميل مع الكاتب والوقت

توصية الذكاء الاصطناعي

اتصال تتطلب مراجعة المدير المالي

لماذا؟

- أقدم فاتورة متأخرة منذ 52 يوماً.
- 310 ألف ج.م متأخرة.
- يدفع عادةً بعد الموعد بـ 32 يوماً.
- تم كسر وعدين من أصل 3.

السياسة

قاعدة مفعّلة: مراجعة المدير بعد 30 يوم تأخير.
سياسة الائتمان §4.2 · تم استرجاع النص لشرحها

الذاكرة المستخدمة

ملاحظة من أحمد، 12 أكتوبر · وعد 24 أكتوبر: مكسور

المسودة (بالعربية) النبرة: رسمية ▾ ودّية · حازمة

السلام عليكم ورحمة الله، السادة شركة ABC للتجارة المحترمين. نود تذكيركم بوجود فواتير مستحقة السداد بإجمالي 310,000 جنيه، وأقدمها منذ 52 يوماً. نرجو التكرم بإفادتنا بموعد السداد المناسب خلال هذا الأسبوع حتى نتمكن من ترتيب الأمر. ولكم جزيل الشكر.

✓ اجتازت المسودة الفحص: لا تهديدات قانونية ولا عبارات ممنوعة.

اعتماد | تعديل | رفض

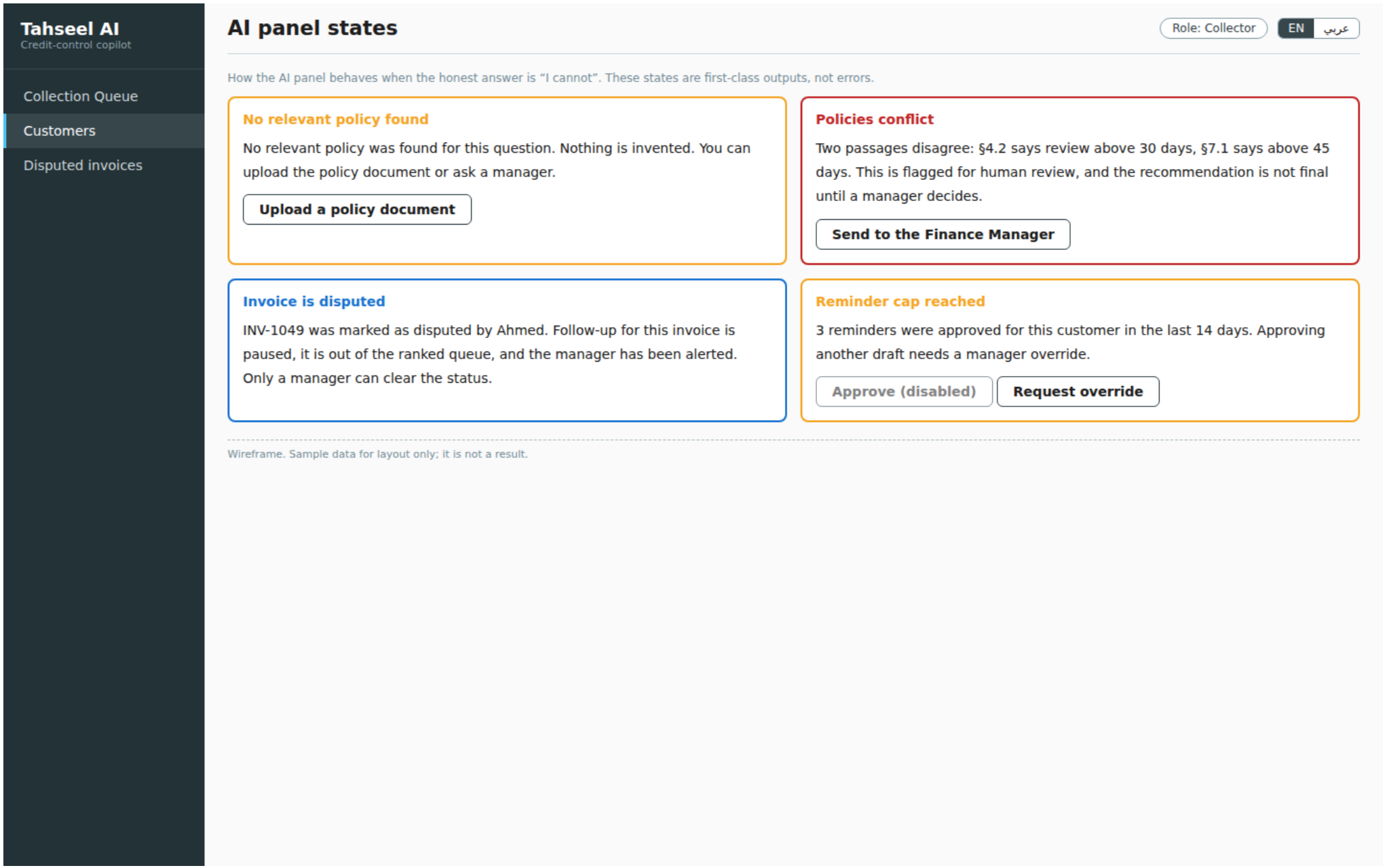
تحصيل لا يرسل الرسالة أبداً؛ يرسلها شخص. للمحصّل يتحول زر الاعتماد إلى «إرسال لمراجعة المدير».

اسأل الوكيل عن هذا العميل...

هيكل تخطيطي. بيانات تجريبية لتوضيح التصميم فقط وليست نتيجة.

4.5 AI panel states

The panel is designed for the cases where the honest answer is "I cannot": no relevant policy, conflicting policies, a disputed invoice, and a reached reminder cap. These are the behaviors that make the system trustworthy, so they get their own designs.


Tahseel AI
Credit-control copilot
Collection Queue
Customers
Disputed invoices
AI panel states
Role: Collector
EN
عربي
How the AI panel behaves when the honest answer is "I cannot". These states are first-class outputs, not errors.
No relevant policy found
No relevant policy was found for this question. Nothing is invented. You can upload the policy document or ask a manager.
Upload a policy document
Policies conflict
Two passages disagree: §4.2 says review above 30 days, §7.1 says above 45 days. This is flagged for human review, and the recommendation is not final until a manager decides.
Send to the Finance Manager
Invoice is disputed
INV-1049 was marked as disputed by Ahmed. Follow-up for this invoice is paused, it is out of the ranked queue, and the manager has been alerted. Only a manager can clear the status.
Reminder cap reached
3 reminders were approved for this customer in the last 14 days. Approving another draft needs a manager override.
Approve (disabled)
Request override
Wireframe. Sample data for layout only; it is not a result.

حالات لوحة الذكاء الاصطناعي

عربي | EN — الدور: محصّل

كيف تتصرف لوحة الذكاء الاصطناعي حين تكون الإجابة الصادقة «لا أستطيع». هذه الحالات مخرجات أساسية وليست أخطاء.

لا توجد سياسة ذات صلة

لم يتم العثور على سياسة ذات صلة بهذا السؤال. لا يتم اختلاق أي شيء. يمكنك رفع وثيقة السياسة أو سؤال المدير.

[رفع وثيقة سياسة]

تعارض في السياسات

نصّان متعارضان: البند §4.2 يقول مراجعة بعد 30 يوماً، والبند §7.1 يقول بعد 45 يوماً. تم تعليم الحالة للمراجعة البشرية، والتوصية ليست نهائية حتى يقرر المدير.

[إرسال إلى المدير المالي]

الفاتورة محل نزاع

تم تعليم الفاتورة INV-1049 كمحل نزاع بواسطة أحمد. المتابعة لهذه الفاتورة متوقفة، وخرجت من القائمة المرتبة، وتم تنبيه المدير. المدير فقط يمكنه إزالة الحالة.

تم بلوغ حد التذكيرات

تم اعتماد 3 تذكيرات لهذا العميل خلال آخر 14 يوماً. اعتماد مسودة أخرى يتطلب موافقة استثنائية من المدير.

[اعتماد (معطّل)] [طلب استثناء]

هيكل تخطيطي. بيانات تجريبية لتوضيح التصميم فقط وليست نتيجة.

5. How the screens connect to the system

Screen element	Comes from	Capability	Who sees it
Totals, aging, days beyond terms, DSO	Receivables engine (`get_portfolio_kpis`)	C2, S3	Owner, Finance Manager
Expected cash with range and confidence	Forecast (`forecast_cash`)	S1	Owner, Finance Manager
Risk score and factors	`calculate_risk`	C3	All roles (Collector: assigned customers)
Priority and suggested action	`calculate_priority` and the suggested-action rules	C4	All roles
Policy rule and cited passage	`retrieve_policy` (rules by code, passages by retrieval)	C5, S5	All roles
Memory used	`get_customer_history` (customer memory, read only)	C6	All roles
Record a call result and a promise	Written by a person through the API	C7	All roles (Collector: assigned customers)
Arabic draft, tone, draft check	`draft_message` and the rule-based check	C9, S7	All roles
Approve, Edit, Reject	The approval gateway at the API	C10	Collector for routine drafts; a manager for flagged ones
Disputed status	Set by a person; the system then pauses follow-up	S7	Set: all roles on assigned customers; cleared: managers
Weights, caps, tone defaults, rules	`tenant_settings`, `policy_rules`	S5, S6	Managers

6. Screen states

State	What the user sees
Loading	A skeleton of the layout; the agent shows which tool it is calling
Empty queue	"No overdue accounts today", with the date of the last import
Import problems	Row-level warnings and errors with fix or skip choices
No relevant policy	The panel says so and offers to upload a policy document
Policies conflict	The panel shows both passages and sends the case to a manager
Disputed invoice	The invoice leaves the ranking; follow-up is paused; the manager is alerted
Reminder cap reached	Approve is disabled and a manager override can be requested
Not permitted	A short message names the role that can do it
Error	A plain message and a retry; nothing is half-saved

7. Bilingual and accessibility rules

- **Right to left.** The Arabic UI mirrors the layout. The code uses CSS logical properties, so one stylesheet serves both directions. Direction-bearing icons such as arrows are mirrored; others are not.
- **Digits and formats.** Western digits by default. Amounts and dates are formatted with `Intl`, as "EGP 310,000" in English and "310,000 ج.م" in Arabic.
- **Mixed text.** Invoice numbers, customer names in Latin letters, and codes keep their own direction inside Arabic sentences.
- **No hard-coded text.** Every string comes from a translation dictionary, from the first screen built.
- **Color is never the only signal.** Risk levels show a word and a number as well as a color. Warnings carry an icon or a label.
- **Contrast and keyboard.** Text meets the WCAG AA contrast ratio, every action is reachable by keyboard, and focus is visible.
- **The draft is Arabic.** Message drafts default to Arabic in both interface languages, with tone options: formal, friendly, and firm. The default tone follows the customer tier.

8. Visual style (to confirm)

The wireframes are grayscale with one accent color. The final style is decided when the interface is built:

Item	Working choice
Color	A neutral base, one primary blue, and four risk colors with labels
Type	A font family with good Arabic and Latin coverage, chosen after testing
Spacing	An 8-pixel grid
Layout	Desktop first (wireframes at 1440 pixels wide); tablets next; phones are not part of the MVP

9. Limits of this design

- The wireframes are low fidelity and have not been tested with real users.
- The Arabic text was written for layout and needs review by native speakers on the team before release [status to confirm].
- All numbers are sample data. The demo will show values from the data generator.
- The real font and visual style are not chosen yet.
- Which Should-tier screens exist in the final build depends on the cut order in the Decision Log.